

An Explainable AI Approach for Interpreting Regionally Optimized Deep Neural Networks in Hydrological Prediction

F. Hosseini ^a, C. Prieto ^a, C. Álvarez ^a

^a Instituto de Hidráulica Ambiental de la Universidad de Cantabria, Santander, Spain

Abstract

The interpretation of artificial intelligence (AI) and deep learning (DL) model outcomes remains a central challenge in hydrology and rainfall-runoff modeling. This study investigates whether hyperparameter-optimized regional Long Short-Term Memory (LSTM) networks can implicitly learn hydrological processes directly from hydrometeorological data, without access to explicit catchment attributes during training. Specifically, we explore to what extent these models reinforce classical hydrological understanding or reveal new insights through explainable AI (xAI) analyses.

Using hourly precipitation, temperature, and potential evapotranspiration data from 40 humid and flashy catchments in the Basque Country, Spain, we demonstrate that systematically optimized LSTMs exhibit strong generalization and scalability in regional rainfall-runoff modeling. Through a combination of correlation analysis, Random Forest (RF) modeling, Principal Component Analysis (PCA), and SHAP-based feature attribution, we quantify how catchment attributes indirectly influence LSTM performance. This multi-

method approach provides a novel framework to assess the hydrological “learning maturity” of deep neural networks in regional hydrology.

The results show that LSTM networks implicitly capture latent catchment characteristics that shape hydrological responses. Catchments with high runoff coefficients and higher mean annual streamflow tend to yield more accurate predictions, while catchments characterized by steep slopes, extreme flow variability, and high precipitation variability pose greater challenges due to their nonlinear hydrological behavior. SHAP analysis confirms that both catchment properties (e.g., average yearly runoff coefficient, precipitation, streamflow) and key LSTM hyperparameters (e.g., input sequence length, hidden size, dropout rate) play critical roles in predictive success, with the latter influencing the models’ ability to better generalize and capture extremes.

Furthermore, RF and PCA highlight essential factors influencing model accuracy, including annual precipitation, aridity index, stream density, and land cover, where broadleaf forests improve water retention and urbanization complicates runoff processes. These findings bridge the gap between the “black-box” nature of AI/DL models and hydrological interpretability, offering evidence-based guidelines for practitioners and researchers deploying LSTMs in regional hydrological contexts.

Ultimately, this research underscores the potential of optimized LSTM networks to generalize hydrological processes and reinforce domain knowledge, while also advocating for the integration of catchment attributes in future model designs to enhance predictive

39 robustness. By advancing xAI methodologies for deep learning in hydrology, this study
40 contributes to developing more reliable and interpretable AI-driven solutions for water
41 resource management and flood risk assessment under increasing climate variability.

42 **Keywords**

43 Regional rainfall-runoff modeling, deep neural networks, hyperparameter optimization,
44 explainable AI (xAI), catchment attributes.

Highlights

- Optimized LSTMs generalize hydrological processes without explicit catchment attributes.
- RF and PCA identify key climatic and topographic factors affecting LSTMs' accuracy.
- LSTMs perform accurately in wet catchments and excel in hourly prediction.
- Explainable AI enhances hydrological modeling for large-scale water management.

1. Introduction

The rise of Artificial Intelligence (AI) (Russell & Norvig, 2020) and deep learning (DL) (Goodfellow et al., 2016) has revolutionized numerous fields, including hydrology. Traditional hydrological models, such as conceptual and physically-based approaches, rely on explicitly defining relationships between meteorological variables and hydrological responses based on physical processes (Refsgaard et al., 2022; Beven, 2012). These models incorporate catchment attributes, including climate, topography, geology, land use, and vegetation, to predict hydrological processes like runoff and streamflow. However, the increasing availability of Big data (e.g., large hydrometeorological datasets) and advancements in our computational power by Graphics Processing Units (GPUs) have positioned AI/DL models, particularly Deep Neural networks (DNNs) (e.g., Long Short-Term Memory (LSTM) (Hochreiter & Schmidhuber, 1997)), as powerful alternatives for modeling the complex, nonlinear relationships in rainfall-runoff patterns (Tripathy & Mishra, 2024; Arsenault et al., 2023; Kratzert et al., 2024; Shen & Lawson, 2021).

LSTMs excel in sequential data processing and temporal dependency modeling, often outperforming hydrological traditional models in predictive accuracy (Donnelly et al., 2024). Unlike conventional models that explicitly use catchment attributes, LSTMs implicitly learn latent hydrological features from input-output data (Kratzert et al., 2018). However, the black-box nature of these DNNs has raised concerns about interpretability and trustworthiness. A fundamental question remains: *Do DNNs (e.g., LSTMs) merely mimic observed statistical*

patterns, or do they capture and learn the underlying physical processes governing catchment behavior?

Hydrological phenomena are inherently influenced by spatial and temporal variability, requiring models to account for diverse catchment-specific characteristics (Beven, 2020). For DNNs to gain broader acceptance in regional hydrology, they must address two critical aspects: predictability—generating accurate outputs—and understanding—uncovering the physical relationships driving these predictions (Başagaoglu et al., 2022). While conceptual models explicitly define mechanisms of hydrological behavior, DNNs infer relationships solely from data, emphasizing the need for hybrid approaches that integrate the strengths of both paradigms.

Studies on physics-informed data-driven models have sought to train AI frameworks using physics-based information to enhance interpretability (Donnelly et al., 2024; Xie et al., 2021; Hoedt et al., 2021; Samek et al., 2021; Kraft et al., 2020; Reichstein et al., 2019). Others have employed DNNs (e.g., LSTMs) to improve systematic calibration of traditional hydrological models, providing a bridge between these approaches (Tsai et al., 2021). Nonetheless, a significant gap persists in understanding how catchment-specific attributes influence DNNs' performance, particularly in regional contexts. For example, regional LSTM networks trained on multiple catchments data have been shown to outperform single-catchment models by employing shared regional hydrological information (Hosseini et al., 2025; Kratzert et al., 2024). However, the interplay between DL model performance and physical catchment attributes, such as climate, land cover, and soil type, remains underexplored.

Explainable AI (xAI) techniques (Başğaoğlu et al., 2022) have emerged as valuable tools for overcoming the interpretability challenges of machine learning models in hydrology. These methods aim to elucidate relationships between input features and model predictions, shedding light on the latent knowledge embedded through the training process. For example, Lees et al. (2021) demonstrated that LSTMs could replicate hydrological concepts like soil moisture and snow cover storage, with internal memory states showing strong correlations with these variables. Similarly, Kratzert et al. (2018) observed that LSTMs developed specialized memory cells for snow-driven catchments, effectively mimicking the behavior of conceptual snow storage models. These findings underscore the potential of DL models to internalize hydrological processes when trained on well-prepared datasets, moving beyond mere statistical mimicry to capturing underlying physical phenomena.

However, according to Başğaoğlu et al., 2022, challenges remain. Many xAI techniques rely on historical data, limiting their adaptability to nonstationary conditions caused by climate change and human activities. Additionally, their reliability often depends on the quality of available datasets, which can pose difficulties in data-scarce regions. Hybrid models that combine AI techniques with domain knowledge or physics-based constraints offer a potential solution.

Building on these advancements, this study investigates whether regionally optimized LSTM networks (only trained on hydrometeorological data) can capture the physical characteristics of different catchments without explicitly being exposed to unique catchments' attributes. While previous studies emphasize LSTMs' predictive capabilities, their ability to learn hydrological relationships unique to specific catchments has not been systematically examined. Exploring this

ability is essential for bridging the gap between ML efficacy and physical hydrological understanding, especially in regional rainfall-runoff modeling.

We hypothesize that *regionally optimized LSTM networks, trained exclusively on hydrometeorological data while being blind to catchments attributes, are influenced by unique catchment-specific characteristics*. We posit that DNN model performance will vary based on the catchments' attributes and that optimized regional LSTMs can implicitly learn hydrological relationships unique to specific place.

This study addresses the following objectives:

1) Correlation Between LSTM Performance and Catchment Attributes: To what extent do catchment attributes, such as climatic conditions, topography, and land use, influence the performance metrics of optimized regional LSTM networks?

2) Latent Learning of Unique Catchment-Specific Features: Can regional LSTMs trained solely on hydrometeorological data implicitly learn catchment-specific features that affect their prediction accuracy in different places?

3) Impact of Hyperparameter Optimization: How does systematic hyperparameter optimization influence LSTM performance across diverse catchments in regional hydrology?

By exploring these questions, this work aims to enhance the interpretability, reliability, and performance of DNNs, specifically LSTMs, in hydrology, contributing to their application in real-world water management. Furthermore, it seeks to uncover latent hydrological insights embedded in DL models, complementing traditional modeling understandings.

2. Method

2.1. Case Study and Dataset: Basque Country Hydrological System

This study focuses on Basque Country, located in north of Spain along the European Atlantic coast (Figure 1). Covering approximately 4,494 km², the studied region is characterized by its humid climatology, abundant water resources, and diverse hydrological behaviors. The area includes 40 catchments ranging from small basins of 4 km² to large basins of 1,000 km². These catchments, noted for their flashy and humid characteristics, are situated between the Cantabrian Mountains—rising to elevations of 1,300 meters—and the Atlantic Ocean.

The region's landscape consists predominantly of grasslands and evergreen forests, benefiting from the warming influence of the Gulf Stream. The climate is humid and temperate, with mean annual temperatures ranging from 9°C in the mountains to 15°C in lower regions. Annual precipitation varies between 1,200 mm and 1,600 mm, driven primarily by North Atlantic weather fronts. These climatic and geographic conditions make the region prone to intense rainfall events, rapid runoff, and a heightened risk of flash floods, which makes accurate hourly predictions a vital hydrological task in this region.

For this study, detail hydrological and physical attributes of the 40 catchments were gathered (Table 1). The definitions of these attributes can be found in Table 2. This information is essential for understanding the hydrological relationships between unique catchment characteristics and regional model performance in different basins.

The Basque Water Agency (URA), a regional governmental entity, manages water resources and policies in this area. URA has compiled a comprehensive and high-quality dataset of hourly

hydro-meteorological timeseries. In this study we utilized data for over 21 years from October 1, 2000 to September 30, 2021. This dataset underpins the development of advanced hydrological models to enhance water management and flood prediction strategies.

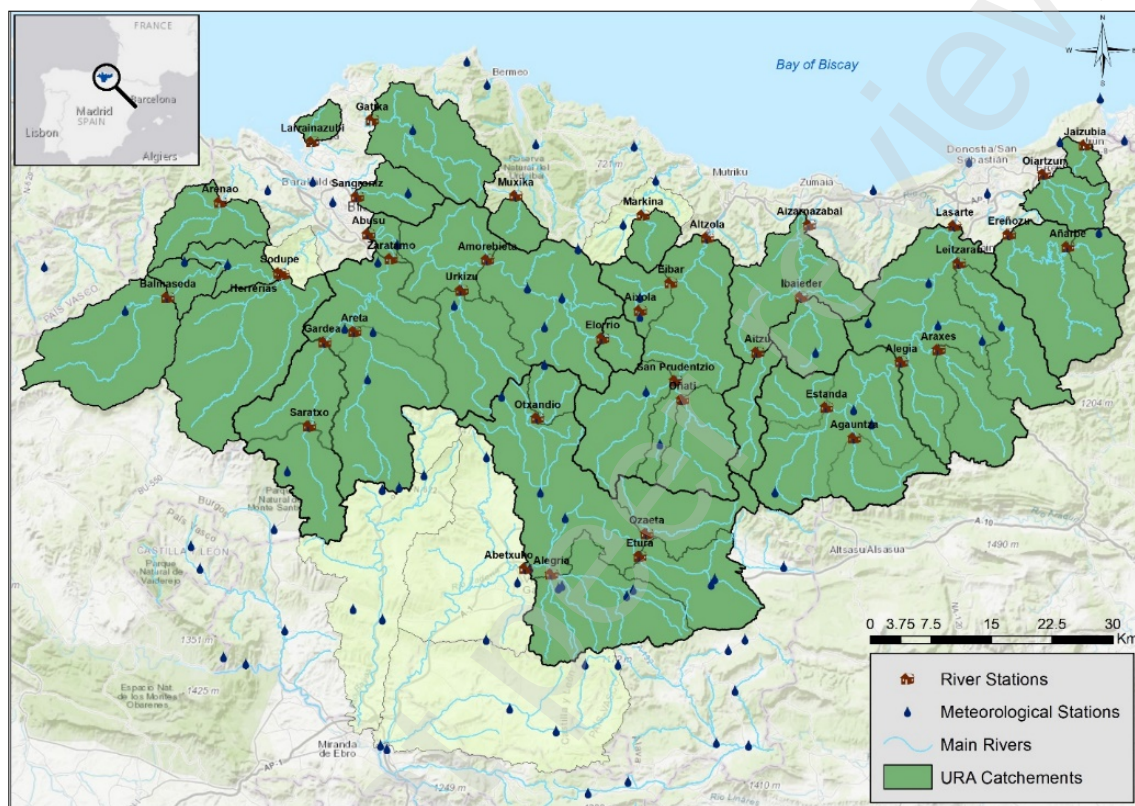


Figure 1. Study area, including 40 catchments of Basque Country in north of Spain

Table 1. A brief summary of 40 URA catchments' attributes in the case study.

Row	Station Name	Data Start	Data End	Area Km2	Annual PREC mm	Min Temp °C	Max Temp °C	Days with Negative Temp	RC	Aridity Index (AI)	Flow mm	Annual PET mm	Elevation			Land Use								Soil				COND PERM HARD				
													GRAD %	MIN Height m	MAX Height m	UHD %	AGR %	PAS %	BLF %	CNF %	PLT %	SSH %	WAE %	DEN %	CALC %	CONG %	SDIM %	VLC %	WATR %	(1-5) class		
1	Abetxuko	9/3/2010	7/14/2019	679.7	1175	-9.91	39.29	225	0.33	1.31	241.31	968.09	25%	503	1549	3%	18%	13%	36%	7%	8%	14%	1%	1%	88%	0%	10%	0%	2%	4.6	3.0	2.8
2	Abusu	10/1/2000	9/30/2021	1003.1	1176	-4.26	38.14	41	0.61	0.82	718.96	956.83	40%	13	1377	2%	0%	18%	18%	24%	13%	23%	0%	3%	97%	2%	0%	0%	1%	4.9	3.1	3.0
3	Agautza	10/1/2000	9/30/2021	69.5	1537	-5.80	38.78	47	0.72	0.77	874.63	937.01	47%	184	1412	0%	0%	25%	50%	4%	13%	8%	0%	1%	100%	0%	0%	0%	0%	5.0	3.1	3.0
4	Aitzu	10/1/2000	9/30/2021	56.8	1486	-5.90	37.57	51	0.63	0.67	886.67	951.77	45%	312	1431	1%	0%	21%	22%	31%	17%	5%	0%	3%	100%	0%	0%	0%	0%	5.0	3.1	3.0
5	Aixola	7/1/1987	9/30/2019	4.8	1421	-17.30	38.13	40	0.44	0.65	619.60	929.90	44%	340	750	0%	0%	5%	5%	80%	10%	0%	0%	0%	100%	0%	0%	0%	0%	4.6	3.0	3.0
6	Aizarnazabal	10/1/2000	9/30/2021	273.5	1653	-5.90	37.57	51	0.63	0.67	895.56	950.02	47%	20	1074	1%	1%	32%	14%	43%	4%	5%	0%	0%	97%	3%	0%	0%	0%	5.0	3.1	3.0
7	Alegia	10/1/2000	10/2/2020	329.6	1416	-6.03	39.24	61	0.61	0.76	770.05	959.35	45%	92	1549	1%	0%	27%	30%	14%	18%	10%	0%	1%	100%	0%	0%	0%	0%	5.0	3.1	3.0
8	Alegria	4/30/2010	2/7/2021	185.1	1074	-10.74	39.45	238	0.31	1.18	231.54	868.46	22%	508	1099	4%	36%	3%	42%	0%	5%	9%	1%	0%	81%	0%	19%	0%	0%	4.3	3.0	2.6
9	Altzola	10/1/2000	9/30/2021	460.3	1380	-5.34	39.28	44	0.60	0.74	775.15	955.84	44%	12	1363	2%	0%	26%	20%	25%	15%	11%	0%	1%	99%	0%	0%	0%	0%	5.0	3.1	3.0
10	Amorebieta	10/1/2000	9/30/2021	233.4	1302	-5.72	40.12	51	0.71	0.73	912.70	950.78	43%	65	1330	2%	1%	24%	10%	27%	11%	17%	0%	8%	90%	10%	0%	0%	0%	4.7	2.9	2.9
11	Anarbe	10/1/2000	9/30/2021	47.1	2031	-7.87	39.63	74	0.78	0.42	1744.02	951.18	54%	182	1052	0%	0%	18%	62%	5%	12%	2%	0%	0%	0%	87%	0%	12%	0%	1.9	2.9	4.0
12	Araxes	1/5/2011	9/30/2021	93.1	1616	-7.00	38.69	69	0.75	0.50	1428.12	950.78	49%	119	1429	1%	0%	30%	41%	13%	11%	3%	0%	1%	100%	0%	0%	0%	0%	5.0	3.3	3.0
13	Arenao	5/26/2005	9/1/2020	85.7	1200	-3.74	39.31	14	0.58	0.87	616.01	929.71	38%	45	821	1%	0%	28%	16%	20%	12%	20%	0%	4%	100%	0%	0%	0%	0%	5.0	3.1	3.0
14	Areta	4/1/2013	9/30/2021	190.1	1149	-5.14	37.88	48	0.58	0.89	625.02	953.53	36%	118	1305	1%	0%	14%	37%	15%	9%	23%	0%	1%	100%	0%	0%	0%	0%	5.0	3.0	3.0
15	Balmaseda	10/1/2000	9/30/2021	194.9	1064	-4.07	40.60	18	0.67	0.92	696.91	953.53	34%	172	1331	1%	3%	33%	36%	10%	7%	8%	0%	1%	90%	10%	0%	0%	0%	4.9	2.9	2.9
16	Eibar	12/1/2013	9/30/2021	50.0	1451	-5.04	38.35	25	0.54	0.66	738.08	903.95	44%	94	812	5%	0%	34%	5%	41%	9%	5%	0%	0%	100%	0%	0%	0%	0%	4.6	3.0	3.0
17	Elorrio	1/1/2001	9/30/2021	29.6	1368	-5.02	42.89	18	0.50	0.69	656.77	909.78	40%	167	1116	2%	1%	21%	4%	31%	21%	20%	0%	1%	100%	0%	0%	0%	0%	5.0	3.0	3.0
18	Erenozu	10/1/2000	9/30/2021	215.8	1902	-6.43	38.56	36	0.72	0.45	1444.70	909.78	55%	23	1142	0%	0%	20%	41%	15%	15%	9%	0%	0%	2%	95%	0%	2%	1%	2.3	2.8	3.7
19	Estanda	10/1/2000	9/30/2021	54.6	1352	-5.80	38.78	47	0.48	0.75	587.45	909.78	43%	164	966	2%	0%	26%	15%	32%	13%	11%	0%	0%	100%	0%	0%	0%	0%	5.0	3.0	3.0
20	Etura	12/1/2012	9/30/2021	113.9	1201	-9.47	35.91	135	0.63	1.02	571.04	924.84	20%	548	1150	3%	37%	14%	39%	0%	1%	4%	0%	0%	90%	3%	7%	0%	0%	4.5	3.0	2.8
21	Gardea	10/1/2000	9/30/2021	192.2	1025	-5.55	41.73	61	0.42	0.94	418.20	939.40	34%	134	1183	1%	1%	26%	27%	15%	8%	21%	0%	1%	98%	2%	0%	0%	0%	5.0	3.0	3.0
22	Gatika	12/1/2013	9/30/2021	143.4	1308	-2.88	37.71	11	0.61	0.77	747.34	940.40	28%	5	687	6%	1%	38%	16%	20%	5%	13%	0%	1%	96%	4%	0%	0%	0%	4.8	3.0	3.0
23	Herrerias	10/1/2000	9/30/2021	254.0	1103	-4.07	40.60	18	0.41	0.84	450.38	926.70	39%	50	1184	0%	0%	26%	14%	27%	13%	19%	0%	1%	100%	0%	0%	0%	0%	5.0	3.0	3.0
24	Ibai Eder	10/1/2000	9/30/2021	65.4	1644	-5.90	37.57	51	0.51	0.62	738.76	895.34	49%	87	971	0%	0%	26%	27%	36%	3%	7%	0%	0%	100%	0%	0%	0%	0%	5.0	3.0	3.0
25	Jaizubia	4/1/2013	9/30/2021	18.3	1988	-3.72	41.03	7	0.82	0.60	1244.23	902.12	36%	3	544	10%	0%	20%	33%	1%	9%	26%	0%	0%	52%	32%	0%	16%	0%	3.3	2.8	3.4
26	Larrainazubi	4/1/2013	9/30/2021	19.1	1363	-3.55	39.57	4	0.74	1.01	702.34	953.67	23%	5	254	20%	0%	35%	29%	0%	0%	15%	0%	0%	12%	86%	2%	0%	0%	3.3	2.1	2.1
27	Lasarte	10/1/2000	9/30/2021	791.3	1601	-5.77	38.84	50	0.69	0.62	971.40	874.21	45%	18	1549	1%	0%	28%	28%	21%	11%	9%	0%	1%	83%	17%	0%	0%	0%	4.7	3.1	3.0
28	Leitzaran	10/1/2000	9/30/2021	114.2	1943	-7.34	38.69	69	0.75	0.47	1490.37	938.93	49%	49	1200	1%	0%	21%	22%	42%	4%	11%	0%	0%	17%	83%	0%	0%	0%	3.2	2.7	3.1
29	Markina	12/1/2013	9/30/2021	34.0	1469	-5.46	37.80	28	0.61	0.71	823.09	955.75	45%	76	791	1%	0%	35%	3%	48%	5%	8%	0%	0%	100%	0%	0%	0%	0%	4.8	3.0	3.0
30	Muxika	10/1/2000	9/30/2021	31.4	1351	-3.56	39.10	37	0.47	0.69	650.36	945.24	36%	11	625	0%	1%	25%	5%	43%	11%	14%	0%	0%	97%	3%	0%	0%	0%	4.2	3.0	3.0
31	Oartzun	10/1/2000	9/30/2020	55.9	2073	-5.49	38.57	27	0.81	0.46	1517.96	860.62	49%	6	831	1%	0%	19%	25%	9%	15%	26%	0%	4%	18%	38%	0%	44%	0%	2.1	2.5	3.8
32	Onati	10/1/2000	9/30/2021	99.1	1442	-5.34	39.28	44	0.63	0.63	997.41	1005.28	47%	193	1362	1%	0%	21%	30%	19%	15%	12%	0%	2%	100%	0%	0%	0%	0%	5.0	3.0	3.0
33	Otxandio	1/1/2003	9/30/2021	35.5	1361	-9.23	38.06	121	0.70	0.71	956.55	962.09	34%	549	1330	1%	0%	11%	40%	29%	7%	8%	0%	5%	100%	0%	0%	0%	0%	5.0	3.0	3.0
34	Ozaeta	4/1/2014	9/30/2021	97.5	1323	-8.58	36.86	90	0.58	1.04	540.12	961.37	29%	549	1547	3%	4%	24%	32%	0%	10%	25%	0%	1%	86%	0%	14%	0%	0%	4.6	3.0	2.7
35	San Prudentzio	10/1/2000	9/30/2021	122.2	1206	-5.34	39.28	44	0.55	0.77	712.99	992.86	42%	171	1146	2%	0%	19%	23%	12%	26%	15%	0%	3%	99%	0%	0%	0%	1%	5.0	3.2	3.0
36	Sangroniz	6/6/2005	9/30/2021	50.8	1281	-4.56	39.87	27	0.50	0.76	598.78	922.30	28%	5	475	11%	2%	26%	18%	16%	6%	19%	0%	1%	100%	0%	0%	0%	0%	4.6	3.0	3.0
37	Saratxo	10/1/2000	9/30/2021	91.2	993	-5.07	41.45	37	0.46	0.94	440.65	910.58	35%	225	1140	1%	1%	27%	40%	5%	1%	23%	0%	2%	94%	6%	0%	0%	0%	4.9	2.9	2.9
38	Sodupe	2/22/2001	8/29/2020	275.8	1118	-4.07	40.60	18	0.68	0.87	710.17	916.49	38%	49	717	1%	0%	27%	10%	28%	15%	19%	0%	0%	99%	1%	0%	0%	0%	5.0	3.0	3.0
39	Urkiu	10/1/2000	9/30/2021	127.1	1178	-5.16	41.29	39	1.02	0.71	1311.44	912.66	42%	68	1377	1%	0%	16%	13%	31%	12%	23%	0%	4%	100%	0%	0%	0%	0%	5.0	3.2	3.0
40	Zaratamo	1/1/2003	9/30/2021	512.3	1107	-4.78	39.28	47	0.57	0.88	607.34	940.69	38%	41	1305	1%	0%	17%	26%	19%	11%	24%	0%	1%	98%	1%	0%	0%	1%	5.0	3.1	3.0

* PREC: precipitation; Temp: Temperature; RC: Runoff Coefficient; PET: Potential Evapotranspiration; GRAD: Gradient (Slope)

* Land Use Distribution: Urban (UHD), Agriculture (AGR), Pasture (PAS), Broadleaf Forest (BLF), Coniferous Forest (CNF), Plantation (PLT), Shrublands (SSH), Water bodies (WAE)

* Soil Composition: CALC: calcareous soils; CONG: conglomerate soils; SDIM: sedimentary soils; VLC: volcanic soils; WATR: wetlands and water associated ecosystems

* Soil Composition class: COND (soil conductivity), PERM (permeability), and HARD (hardness)

Table 2. Definitions of the catchments' attributes employed in this study.

Attribute	Definition	Group	Units/Values
Area	Contributing area to the downstream end of the segment	Topography	km ²
CONF_DEN	Number of rivers confluences by catchment area	Topography	Number/km ²
GRADIENT	Mean gradient through the reach (vertical change/horizontal length)	Topography	ratio
max slope	max slope of catchment	Topography	°
mean slope	average slope of catchment	Topography	°
elevation	Average catchment elevation upstream the river reach	Topography	m
min height	min catchment elevation	Topography	m
max height	max catchment elevation	Topography	m
UHD	Surface occupied by urban areas upstream the river reach	Land Uses	%
AGR	Surface occupied by agricultural land upstream the river reach	Land Uses	%
PAS	Surface occupied by pasture upstream the river reach	Land Uses	%
BLF	Surface occupied by broadleaf forest upstream the river reach	Land Uses	%
CNF	Surface occupied by coniferous forest upstream the river reach	Land Uses	%
PLT	Surface occupied by plantations upstream the river reach	Land Uses	%
SSH	Surface occupied by moors, heathland, scrub and shrubs upstream the river reach	Land Uses	%
WAE	Surface occupied by wetlands and water ecosystems upstream the river reach	Land Uses	%
DEN	Surface occupied by denuded areas upstream the river reach	Land Uses	%
calc	Area occupied by calcareous rocks upstream the river reach	Geology	%
cong	Area occupied by conglomerate rocks upstream the river reach	Geology	%
sdim	Area occupied by sedimentary rocks upstream the river reach	Geology	%
vlc	Area occupied by volcanic rocks upstream the river reach	Geology	%
watr	Area occupied by wetlands and water associated ecosystems upstream the river reach	Geology	%
conductivity	Average soil conductivity upstream the river reach (derived from geology variables). Reaches with MN_watr and MN_oth=1 this value have 0 for this field	Geology	Class: 1 - 5
permeability	Average terrain permeability upstream the river reach (derived from geology variables). Reaches with MN_watr and MN_oth=1 this value have 0 for this field	Geology	Class: 1 - 5
rock hardness	Average soil hardness upstream the river reach (derived from geology variables). Reaches with MN_watr and MN_oth=1 this value have 0 for this field	Geology	Class: 1 - 5
no. prec stations	Number of stations participated in calculating lumped prec values for basins	Hydrology	Number
no. temp stations	Number of stations participated in calculating lumped temp values for basins	Hydrology	Number
possible snow	Percentage of number of days with negative temp on total number of days	Hydrology	%
no. days with negative temp	Number of days with negative temp in the dataset	Hydrology	Number
mean runoff coeff.	yearly average runoff coefficient	Hydrology	dimensionless
aridity index	Aridity Index	Hydrology	dimensionless
mean precipitation	yearly average precipitation	Hydrology	mm
mean streamflow	yearly average streamflow	Hydrology	mm
mean temperature	yearly average temperature	Hydrology	°C
min temperature	yearly min temperature	Hydrology	°C
max temperature	yearly max temperature	Hydrology	°C
Coeff. var. Prec	Coefficient of variation of precipitation	Hydrology	dimensionless
Coeff. var. Flow	Coefficient of variation of streamflow	Hydrology	dimensionless
mean PET	average potential evapotranspiration	Hydrology	mm

162

163 - **Hydro-Meteorological Timeseries and Data Splitting**

164 The training dataset includes precipitation, temperature, potential evapotranspiration (PET),
 165 and streamflow. Catchment-scale areal rainfall and temperature were calculated using the
 166 inverse distance weighting (IDW) method (Ly et al., 2013), and PET was estimated using the
 167 Hargreaves formula (Hargreaves & Allen, 2003).

168 To ensure rigorous model training and evaluation, the dataset was partitioned into distinct
 169 subsets:

170 1) Training-and-Validation Set (October 1, 2000, to September 30, 2015):

- 171 • Training Period: October 1, 2005, to September 30, 2015.
- 172 • Validation Period: October 1, 2000, to September 30, 2005.

173 This subset was employed for hyperparameter optimization and deciding on the finally
174 optimized networks.

175 2) Test Set (October 1, 2015, to September 30, 2021): This subset was withheld during
176 training and validation to provide an unbiased evaluation of the optimized models. Once
177 the model is optimized, it is re-trained on the train set and across 10 different random
178 seeds to ensure robustness, as the initial point of training can still influence the final
179 model's performance. This way, every optimized LSTM concluded in 10 retrained final
180 models that is tested on the test set.

181 **2.2. Model and Main Setups**

182 This study employed the Multi-Timescale Long Short-Term Memory (MTS-LSTM) network, an
183 advanced deep learning architecture tailored for hydrological predictions at fine temporal
184 resolutions. The MTS-LSTM model, introduced by Gauch et al. (2021) and implemented via the
185 NeuralHydrology Python library (Kratzert et al., 2022), is designed to address the computational
186 challenges associated with hourly predictions. By utilizing two parallel LSTM networks—one
187 operating at hourly and the other at daily timesteps—the architecture effectively captures both
188 short-term dynamics and long-term trends while expediting the training process, making it
189 particularly suitable for complex hydrological systems like those of the Basque Country.

For the MTS-LSTM implementation, the inputs were hourly average precipitation, temperature, and Potential Evapotranspiration (PET). These three variables comprehensively reflect the region's humid and temperate climate, providing essential insights into interactions between meteorological conditions and hydrological responses. The target was hourly streamflow in all 40 catchments, simultaneously.

The MTS-LSTM models were trained using the comprehensive dataset from 40 catchments, employing the following rigorous workflow:

1. Hyperparameter Optimization by Random Search (See: Hosseini et al., 2024 for details): Conducted to identify the best-performing configurations by 1000 random search iterations in a very broad hyperparameter space. We considered 10 distinct hyperparameters shown in Table 3 and optimized the regional networks. Finally, 84 distinct networks were selected as regionally optimized networks, regarding their overall average performance metrics on the validation set.
2. Re-training optimized networks with 10 Random Seeds: All optimized networks had high values in terms of their overall average accuracy on both Nash–Sutcliffe (NSE) (Nash & Sutcliffe, 1970) and Kling–Gupta (KGE) (Gupta et al., 2009) efficiency metrics for the whole region on the validation set. The optimized models were retrained on 10 different random seeds to ensure robustness against initialization variability.
3. Evaluation: The final retrained networks were assessed using unseen test data to gauge their generalization capabilities in every catchment.

The performance of the optimized LSTM networks in this research was rigorously evaluated against observed data from the test set. To assess the accuracy of the models predictions, we employed 14 different performance metrics, including Nash-Sutcliffe Efficiency (NSE) (Nash & Sutcliffe, 1970), Kling-Gupta Efficiency (KGE) (Gupta et al., 2009), Mean Squared Error (MSE) (Legates and McCabe, 1999; Makridakis et al., 1993), Root Mean Squared Error (RMSE) (Willmott and Matsuura, 2006), Alpha-Nash-Sutcliffe Efficiency (Alpha-NSE), Beta-Nash-Sutcliffe Efficiency (Beta-NSE), Beta-Kling-Gupta Efficiency (Beta-KGE) (Gupta et al., 2012), Pearson's Correlation Coefficient (Pearson-r), three common biases: High-segment volume (%BiasFHV), Low-segment volume (%BiasFLV), Mid-segment slope (%BiasFMS) (Yilmaz et al., 2008), and three metrics presented by Kratzert et al. (2019, 2020): Mean difference in peak flow timing (Peak-Timing), Mean Absolute Percentage Error for peaks (MAPE_{peak}), and Fraction of Missed Peaks (missed_{peaks}). These metrics, providing different assessment perspectives, result in a complementary approach to thoroughly evaluate the models' performance from various aspects, ensuring a comprehensive evaluation of their predictive accuracy, bias, error distribution, and hydrological behavior.

Moreover, to ensure that regional hydrological behaviors were comprehensively learned, the LSTM networks employed in this analysis were regional in design (meaning that they were trained regionally on data from all 40 catchments). This regional training approach enabled the models to predict streamflow at the outlets of all individual catchments while uncovering insights from the broader dataset. This design choice reflects the study's commitment to developing a scalable

and effective framework for regional rainfall-runoff modeling, highlighting the importance of integrating shared and individual catchment dynamics into the predictive process.

Table 3. The initial defined hyperparameters space designed for random search

Hyperparameter		Range
hidden size		16, 32, 64, 128, 256
batch size		32, 64, 128, 256
output dropout		0, 0.2, 0.4
initial forget bias		-3, -1, 0, 1, 3
learning rates	Lr0	1e-3, 1e-2, 5e-2
	Lr10	5e-4, 1e-3, 5e-3
	Lr25	1e-4, 1e-3
target noise std		0, 0.01, 0.02, 0.05, 0.1
loss function		NSE, RMSE
seq length daily		146, 182, 365, 730, 1095
seq length hourly		168, 336, 504, 672, 1344, 2016, 4032, 6720, 8064, 8760
regularization		tie_frequencies, None

2.3. Test_DATASET Setup and Compilation

To systematically evaluate the implicit learning of catchments' attributes by the optimized deep neural networks in regional rainfall-runoff modeling, we developed a novel method based on their test performances in different basins. Our primary objective was to determine if the optimized regional LSTMs learned some relevant information about catchments' attributes

without direct access to this information during training. Figure 2 provides an overview of the developed methodology for assessing the hydrological understanding of regionally optimized LSTMs.

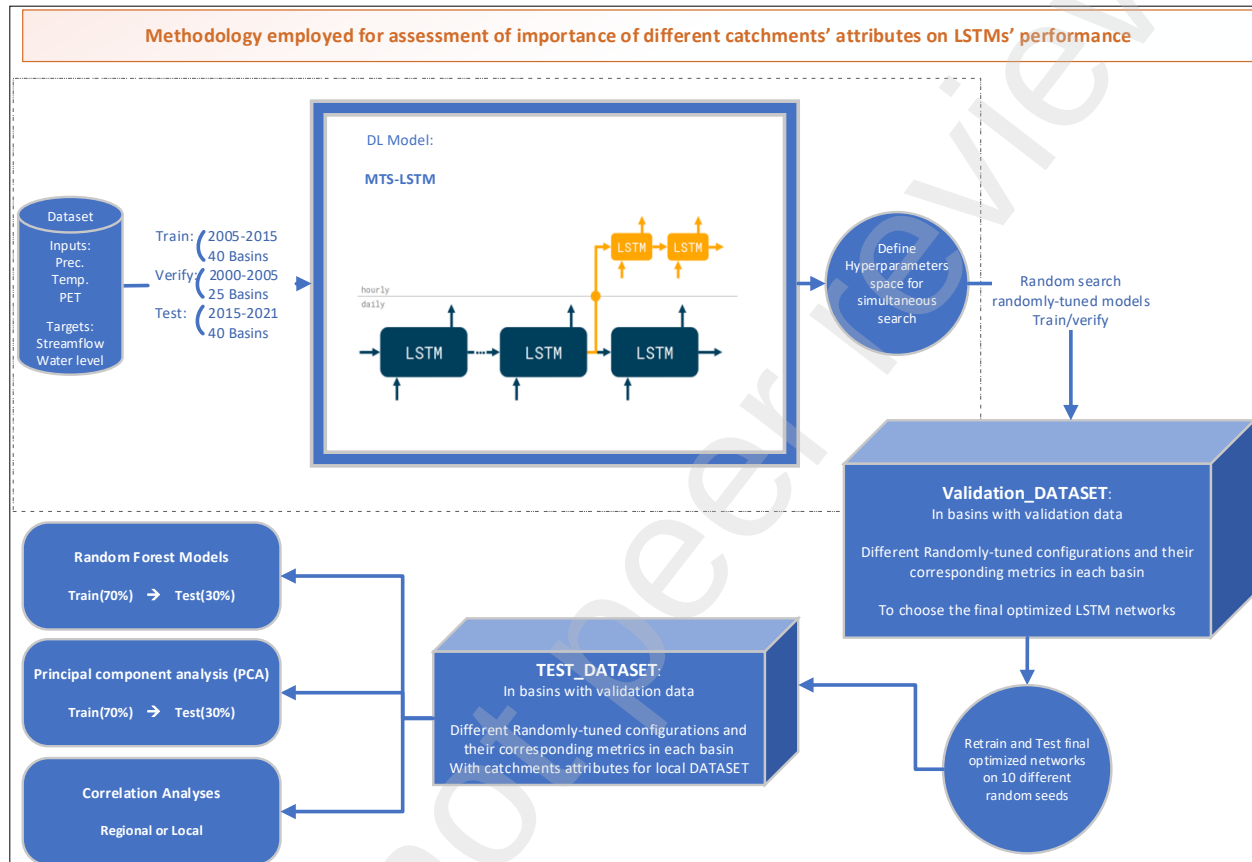


Figure 2. Methodology for Assessing relations between catchments' attributes and LSTMs' test performances

To achieve the objectives, this paper employed a systematic approach to mine and comprehensively analyze what is referred to as the “test_DATASET (with capital letters).” The test_DATASET consists of streamflow test performance metrics from 84 distinct (in terms of their hyperparameter configurations) regionally optimized LSTM networks for rainfall-runoff modeling and catchment attributes across the 40 studied catchments in Basque Country, Spain. All in all, we analyzed performance of 840 trained LSTMs (84 networks retrained on 10 different random

seeds) to study possible relations between catchments attributes and optimized DLs' test performances in different basins.

The test_DATASET consisted of three main group columns:

- 1) Hyperparameter Configurations: A detailed record of the hyperparameter configurations for all 84 optimized LSTM networks.
- 2) Catchments attributes (See: Tables 1 and 2).
- 3) Test Performance Metrics in every catchment: The performance of the 840 optimized LSTM networks in this research was evaluated against observed data from the test set.

Overall, the test_DATASET included up to 67,000 records (comprising hourly and daily predictions on ten different random seeds by 84 distinct hyperparameter-optimized LSTM architectures). This compilation provided a solid foundation for exploring potential correlations and trends between the predictive performance of optimized regional LSTM networks and catchment attributes in the studied region. These records were derived from several models optimized following an exhaustive random search in the hyperparameter space. Each regionally optimized LSTM network in the test_DATASET demonstrated competitive regional accuracy on overall, with some marginal differences. While, all the configured networks exhibited statistically significant varying performance across different locations (See: Hosseini et al., 2024a and 2025 for more details).

2.4. Exploration of Implicit Learning of Catchment Attributes

We developed a triple-confirmation framework to evaluate whether the optimized regional LSTMs implicitly learned unique catchment characteristics, despite not being explicitly exposed

to these attributes during optimization and retraining. This framework integrates Pearson correlation analysis, Random Forest regression (RF), and Principal Component Analysis (PCA) to enhance the explainability of deep neural networks in rainfall-runoff modeling, aligning with contemporary Explainable AI (xAI) methodologies in machine learning (Başğaoğlu et al., 2022). The combined use of these three complementary approaches strengthens the reliability and interpretability of the findings, providing a robust assessment of implicit learning in regional rainfall-runoff modeling by DLs. The combined insights of the triple-confirmation framework, offer valuable guidance for hydrologists when optimizing LSTM networks for regional hydrological predictions.

2.4.1. Pearson Correlation Analyses

To explore potential relationships between unseen catchment attributes and the performance of the optimized regional LSTM models, we conducted a comprehensive Pearson correlation analysis. Heatmaps were employed to visualize these relationships, illustrating the degree of association between performance metrics (e.g., NSE and KGE) and catchment attributes (e.g., area, slope, soil type, and land cover) across all 40 catchments. Each cell in the heatmap quantified the strength of correlation, offering an intuitive overview of how specific physical and hydrological catchment characteristics may influence the predictive accuracy of the LSTM models. This visual representation served as an initial step toward identifying trends and patterns that could reveal the underlying connections between model's learning maturity regarding their test performance in different places and catchment-specific attributes.

To gain deeper insights, we focused on the attributes exhibiting the highest correlation values within the heatmaps (a threshold of 0.3 was considered). This targeted analysis allowed us to pinpoint the most influential catchment features and evaluate their potential role in shaping LSTM' hydrological understandings and final catchment-wise performance. By identifying key attribute-performance relationships, we sought to determine where LSTM models achieved higher predictive accuracy and to provide plausible physical explanations for these observations. This approach aligns with established hydrological principles and emphasizes the importance of considering the "uniqueness of the place" paradigm (Beven, 2020; Beven, 2000) in rainfall-runoff modeling by deep learning.

2.4.2. Catchment-Aware Random Forest Model for Implicit Learning Analysis

To quantify the influence of catchment attributes and hyperparameter configurations on the performance of optimized LSTM models, we employed a Random Forest Regression (RF) model (Breiman, 2001) trained on the test_DATASET. RF models are well-suited for this task due to their ability to handle complex interactions between variables and assess feature importance using multiple interpretability techniques.

Two complementary methods were used to analyze feature importance in the RF model: 1) *Gini gain scores*, which measure how much a feature reduces impurity in the tree-based model (Breiman, 2001; Louppe et al., 2013). 2) SHAP (SHapley Additive exPlanations) values (Eun Choi et al., 2024), which provide an interpretable, game-theoretic approach to feature attribution (Lundberg & Lee, 2017). Gini gain scores indicate how much each feature contributes to improving prediction accuracy by evaluating its role in decision splits within the RF model.

Meanwhile, SHAP values offer a more nuanced understanding by quantifying the exact contribution of each feature to a given prediction, allowing us to distinguish whether a feature positively or negatively impacts the LSTM's performance in different catchments.

We trained an RF model using 70% of the test_DATASET, enabling the model to learn intricate relationships between hyperparameter settings of the LSTM networks, unique catchment attributes, and the corresponding test performance metrics (e.g., NSE, KGE) for each basin. The remaining 30% of the test_DATASET was reserved for validation, ensuring the RF model could generalize to unseen hyperparameter configurations and predict the likely performance of LSTMs in different catchments with their unique features.

In simpler terms, the trained RF model serves as a meta-model that predicts the expected accuracy of a regional LSTM configuration for a specific catchment based solely on its hyperparameter settings and catchment characteristics.

Once trained, the Gini gain scores and SHAP values were analyzed to determine: 1) Key Hyperparameters – Identifying the most critical LSTM hyperparameters influencing model accuracy across all study catchments. 2) Influential Catchment Attributes – Determining the catchment attributes that most significantly impacted the predictive accuracy of regional LSTMs, despite not being explicitly used in the LSTM training process. In other words, the optimized LSTMs, implicitly, learned catchments' uniqueness from latent features in the regional hydrometeorological data of all catchments.

By combining Gini gain and SHAP, we obtained a reinforced interpretability framework that provided deeper insights into the implicit learning capacity of optimized deep neural networks

(LSTMs in our case). This dual analysis allowed us to assess not only which factors were most influential but also the direction and magnitude of their impact on model performance.

This approach strengthens the case for explainable AI (xAI) in hydrology, revealing the extent to which deep learning models—trained without explicit catchment information—implicitly learn hydrological characteristics from latent information buried in regional hydrometeorological data.

2.4.3. Principal Component Analysis (PCA) for Dimensionality Reduction

To complement the RF findings, we employed Principal Component Analysis (PCA) to further explore the relationships between hyperparameters, catchment attributes, and LSTM model performance. PCA is a statistical technique that reduces data dimensionality by transforming the dataset into orthogonal components, capturing the most significant variance. It is widely used for dimensionality reduction, feature extraction, and data visualization (Prieto et al., 2019; 2021; 2022; Jolliffe, 2002).

In this study, PCA served to highlight the influence of hyperparameters and catchment attributes on the performance of regional LSTM models. By analyzing the test_DATASET, PCA allowed us to assess which features contributed most to model performance while revealing latent patterns in the data. This approach also facilitated visualization of complex relationships, aiding in the interpretation of how catchment-specific attributes—such as topography, climate, and land use—affect regional rainfall-runoff modeling by DNNs.

We focused on the first 10 principal components, which explained the majority of variance in the test_DATASET. To enhance interpretability, we generated biplots for the first two principal

components. These biplots provided a visual representation of feature contributions, showing how catchment attributes and hyperparameter configurations influenced predictive accuracy.

This analysis offered critical insights into the implicit learning capabilities of LSTM models trained on regional hydrometeorological data without direct access to catchment attributes. By uncovering the characteristics that most strongly influenced model performance, PCA refined our understanding of the interplay between physical hydrological attributes and optimized LSTM configurations in regional hydrology.

3. Results

We implemented a triple-confirmation analytical approach to investigate the relationships between catchment attributes and the accuracy metrics of optimized regional LSTM configurations on the test set. This approach encompassed Pearson correlation analysis, Random Forest (RF) regression modeling, and Principal Component Analysis (PCA). Together, these methods allowed us to: 1) Examine the associations between catchment attributes and model performance metrics, 2) Evaluate the predictability of LSTM performance using RF models, and 3) Identify the most influential features among the catchments' attributes and hyperparameters.

3.1. Pearson Correlation Analysis

The correlation analysis provided valuable insights into the relationships between specific catchment features (e.g., climate, land use, elevation, and drainage area) and the performance metrics of optimized regional LSTMs across various locations. This analysis helped infer the hydrological understanding encoded in the deep neural networks (DNNs) trained on the Basque Country dataset.

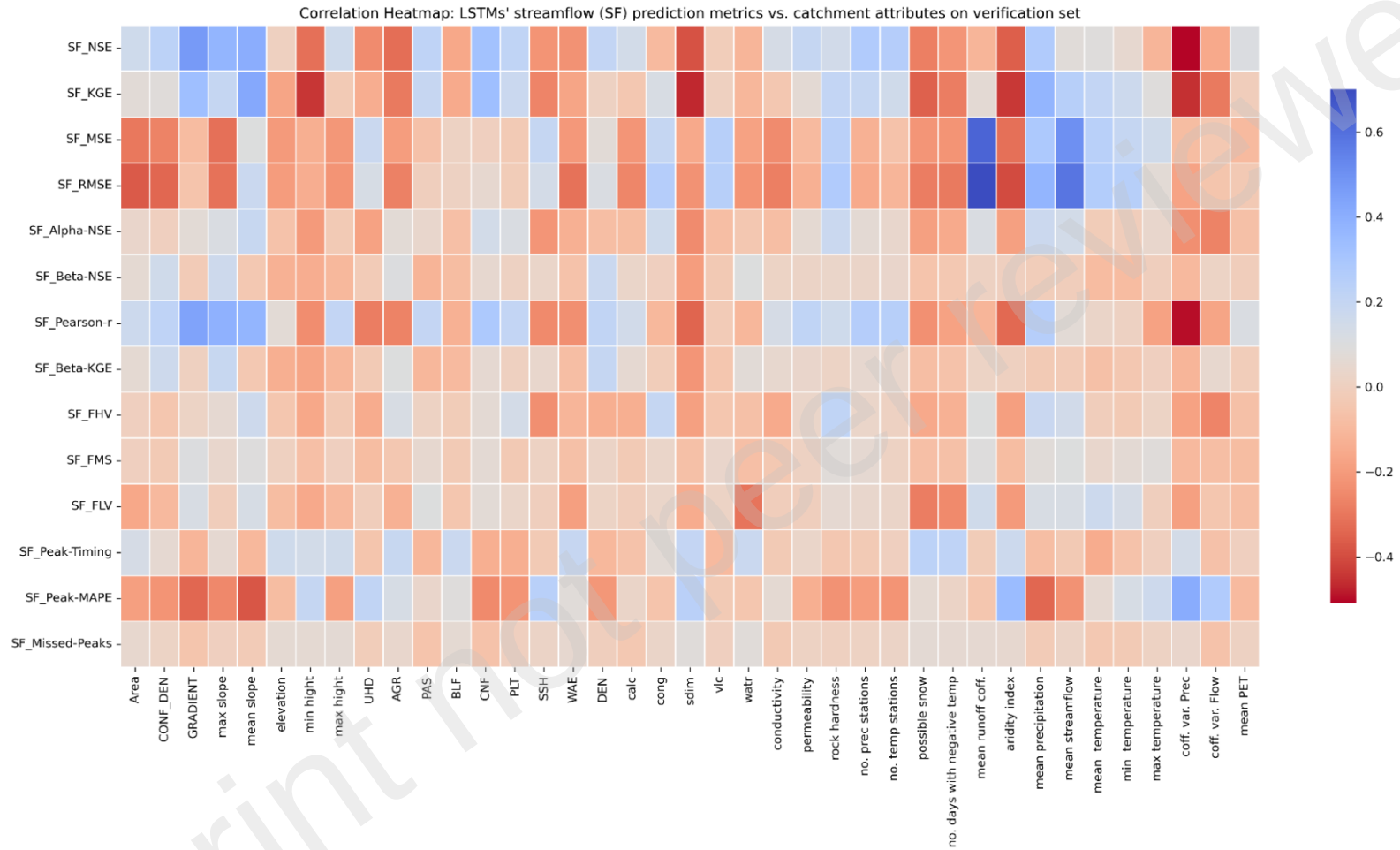


Figure 3. Correlation heatmap showing the relationships between catchment attributes and streamflow (SF) test metrics. Darker colors represent stronger correlations, with both positive and negative relationships between various features and metrics.

Figure 3 presents a Pearson correlation heatmap, illustrating the relationships between catchment attributes and performance metrics for multiple optimized LSTM configurations. Although the LSTM models were trained without explicit knowledge of catchment attributes, they implicitly learned these latent features from information in the lumped input features such as precipitation, temperature, potential evapotranspiration, and streamflow as the target variable for all 40 catchments.

We focused on correlations exceeding ± 0.3 threshold, as presented in Table 4, to identify attributes with meaningful influence on LSTM performance. These correlations revealed trends between catchment attributes and specific metrics, providing a clearer understanding of the conditions under which DNNs performed more accurately or struggled.

Runoff Coefficient and Yearly Streamflow Impact on Prediction Accuracy: High positive correlations between MSE and RMSE test metrics and the average runoff coefficient (0.63 and 0.70, respectively) as well as mean yearly streamflow (0.50 and 0.58, respectively) suggest that catchments with higher runoff coefficients tend to have a bit larger absolute errors in these metrics. Since MSE and RMSE are not dimensionless and scale with streamflow magnitude, larger streamflow naturally leads to higher absolute error values. However, this does not necessarily indicate lower relative predictive accuracy, as normalized performance metrics (e.g., NSE or KGE) may provide a different perspective on model effectiveness.

Given that all models in this study are optimized and demonstrate acceptable accuracy, this suggests that catchments with high runoff coefficients—such as those in the Basque Country—exhibit clear hydrological signals, making flow dynamics easier for LSTM networks to capture.

However, at both very high and very low runoff coefficient values, LSTMs may struggle to accurately learn the trends, potentially leading to slightly diminished performance.

Topographic Influence on Predictive Metrics: Moderate correlations were observed between catchment slope and gradient and metrics like NSE (0.40), KGE (0.42), and Pearson-r (0.44). Steeper catchments likely exhibit distinct runoff patterns with reduced surface retention, aiding DNN model accuracy. However, negative correlations with extreme slopes (e.g., maximum slope attribute: -0.31 for RMSE) suggest that highly steep terrains might introduce noise, complicating model learning due to complex flow dynamics or input data limitations.

Table 4. The attributes with correlations beyond the ± 0.3 threshold, linking them to their influence on LSTM performance.

Metric	Attribute	Correlation	Metric	Attribute	Correlation
SF_RMSE	mean runoff coff.	0.70	SF_MSE	max slope	-0.32
SF_MSE	mean runoff coff.	0.63	SF_NSE	min high	-0.32
SF_RMSE	mean streamflow	0.58	SF_MSE	aridity index	-0.32
SF_MSE	mean streamflow	0.50	SF_NSE	AGR	-0.32
SF_NSE	GRADIENT	0.47	SF_Pearson-r	aridity index	-0.33
SF_Pearson-r	GRADIENT	0.44	SF_RMSE	CONF_DEN	-0.34
SF_KGE	mean slope	0.42	SF_Peak-MAPE	mean precipitation	-0.34
SF_Peak-MAPE	coff. var. Prec	0.41	SF_Peak-MAPE	GRADIENT	-0.34
SF_NSE	mean slope	0.40	SF_Pearson-r	sdim	-0.34
SF_Pearson-r	max slope	0.39	SF_KGE	possible snow	-0.35
SF_Pearson-r	mean slope	0.38	SF_NSE	aridity index	-0.36
SF_KGE	mean precipitation	0.38	SF_RMSE	Area	-0.37
SF_NSE	max slope	0.38	SF_Peak-MAPE	mean slope	-0.38
SF_RMSE	mean precipitation	0.38	SF_NSE	sdim	-0.39
SF_Peak-MAPE	aridity index	0.35	SF_RMSE	aridity index	-0.40
SF_KGE	GRADIENT	0.34	SF_KGE	aridity index	-0.44
SF_KGE	CNF	0.33	SF_KGE	min high	-0.45
SF_NSE	CNF	0.32	SF_KGE	coff. var. Prec	-0.45
SF_FLV	watr	-0.31	SF_KGE	sdim	-0.47
SF_RMSE	max slope	-0.31	SF_Pearson-r	coff. var. Prec	-0.49
SF_RMSE	WAE	-0.31	SF_NSE	coff. var. Prec	-0.51

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Influence of Climate and Variability: The aridity index showed negative correlations with KGE

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(-0.44) and RMSE (-0.40), indicating that less wet catchments with variable precipitation pose

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challenges for the LSTM models. Such variability reduces the generalizability of hydrological

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responses, impacting DLs' prediction accuracy. Similarly, the coefficient of variation for

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precipitation negatively correlated with KGE (-0.45) and Pearson-r (-0.49), emphasizing that

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stable precipitation regimes facilitate better model generalization and accuracy, whereas

414

irregular patterns increase complexity.

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Land Use and Vegetation Cover: Moderate positive correlations between coniferous forest

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cover and metrics like KGE (0.33) and NSE (0.32) suggest that consistent vegetation patterns may

stabilize hydrological responses, improving prediction accuracy. Conversely, agricultural land use showed a negative correlation with NSE (-0.32), likely due to human-induced variability, such as irrigation and land management practices, which disrupt natural flow dynamics.

Geological Characteristics: Negative correlations between sedimentary soils and metrics like NSE (-0.39) and KGE (-0.47) suggest that heterogeneous permeability and storage properties introduce variability, complicating model performance. Similarly, attributes like wetlands (AWE) and water bodies (Watr) correlated positively with FLV (0.31) but negatively with RMSE (-0.31), indicating their role in modulating flow dynamics over time.

Probability of Snowfall and Temperature Variability: The negative correlation of KGE with snowfall probability (-0.35) highlights the challenge of capturing snowmelt dynamics, which introduces delays and variability in runoff. While snowfall is less predominant in Basque Country, its presence in specific basins underscores the complexity of hydrological modeling under such conditions by DNNs. Additionally, temperature variability negatively correlated with NSE, further highlighting the difficulty of handling highly fluctuating thermal conditions.

3.2. Random Forest Analysis

As stated, to further investigate the factors influencing the performance of optimized LSTM models in regional hydrology, we employed a Random Forest (RF) regression approach. The RF model utilized hyperparameter configurations and catchment attributes as inputs, while performance metrics of the LSTM networks served as the output targets. The dataset was divided into 70% for training and 30% for validation, ensuring robust model evaluation.

Figure 4 displays the validation results of the RF model, illustrating the predicted versus actual performance metrics for streamflow across various catchments. The alignment of data points along the 1:1 line, combined with well-accepted metrics such as near-zero mean squared error (MSE) and R-squared values up to 0.97, demonstrates the trained RF model's capability to predict specific LSTM configurations' performance metrics accurately in different places. This indicates that the trained RF model effectively captured the relationships between catchment attributes, hyperparameter configurations, and the resulting LSTM model performance, underscoring its utility in analyzing complex, nonlinear interactions.

3.2.1. Feature Importance Analysis by Gini gains

Feature importance rankings derived from the RF model, presented in Figure 5, reveal the most influential catchment attributes and hyperparameters contributing to regional LSTM model performance in different locations. The Gini gains, used to rank feature importance, highlighted several key findings:

- 1) **Hydrological Attributes:** Attributes related to hydrological processes—such as mean yearly streamflow, precipitation patterns, and the aridity index—emerged as critical predictors for several metrics, including KGE, Beta-NSE, and Missed-Peaks. For instance: Mean yearly streamflow ranked among the most significant predictors for RMSE and Beta-KGE, highlighting its importance in refining the predictive accuracy of LSTM models. Or, aridity index, which reflects climatic conditions, was also influential.

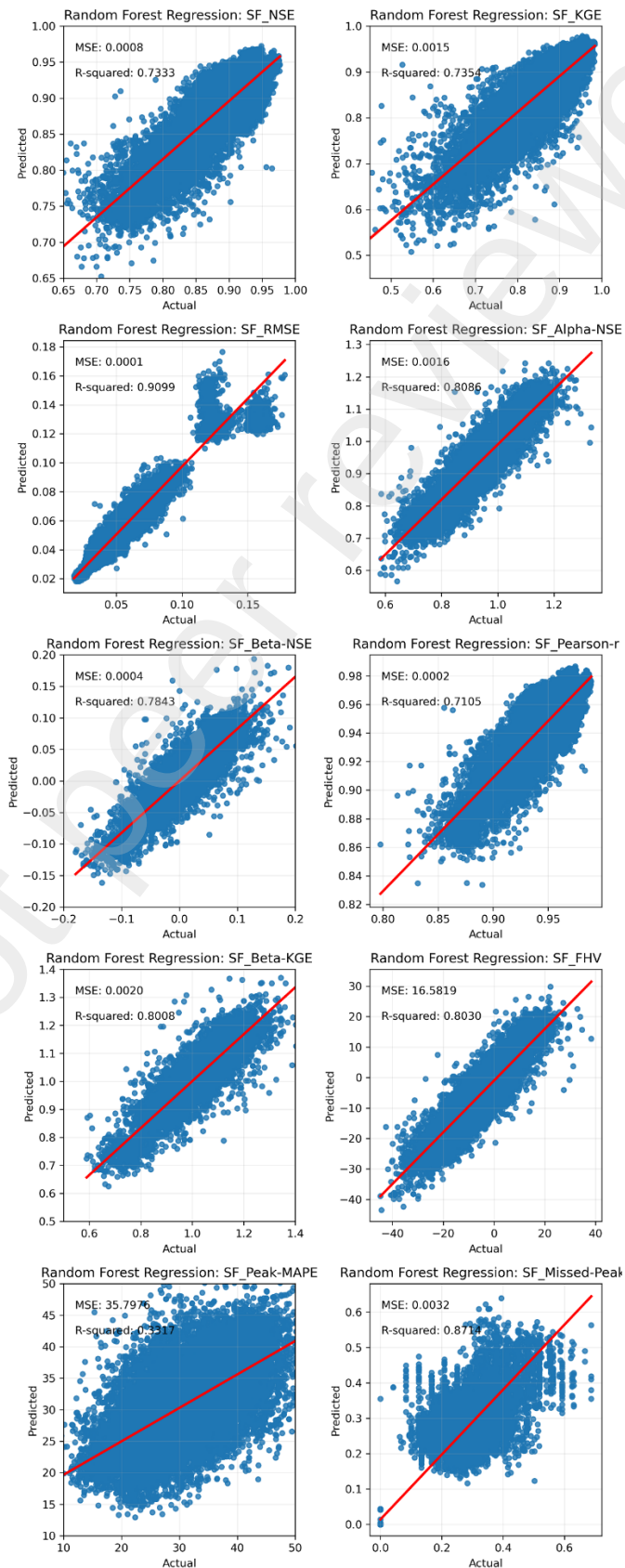
- 2) **Land Cover and Vegetation:** Land cover attributes, such as coniferous forest (CNF) and pasture cover (PAS), played a notable role in influencing accuracy metrics. Specifically,

CNF ranked highly for metrics like NSE, Alpha-NSE, and Pearson-r, suggesting that vegetation types contributing to hydrological consistency can enhance model predictions. In contrast, agricultural land use (AGR) which, also, exhibited negative correlations with performance metrics, demonstrates high Gini values likely due to human-induced variability in hydrological processes.

3) **Topographical and Climatic Factors:** Features such as maximum temperature and maximum slope showed significant importance for specific metrics. Maximum temperature was particularly relevant for metrics like Missed-Peaks, indicating its role in regions with pronounced seasonal variations. Maximum slope showed varying importance, emphasizing the role of topography in shaping hydrological responses.

468

469 *Figure 4. Random Forest prediction*
 470 *accuracy for different test*
 471 *performance metrics for*
 472 *streamflow. The Random Forest was*
 473 *trained on Hyperparameters, and*
 474 *Attributes as inputs and the metrics*
 475 *as outputs. The figure suggests that*
 476 *a Random Forest model can be*
 477 *trained in a way that accurately*
 478 *predicts the performance outcomes*
 479 *of a configured LSTMs in every*
 480 *catchment by knowing their*
 481 *attributes.*



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4) **Hyperparameter Configurations:** Several hyperparameters significantly influenced the RF model's predictive accuracy. However, input sequence length, output dropout, and hidden size were critical for predicting metrics across catchments, from Gini gains perspective, indicating the importance of tailoring model configurations to specific hydrological conditions. Among them, the length of the input sequence could have hydrological importance carrying unique hydrological characteristics of the catchments (Hosseini et al., 2024b) in the hydrological domain. Seed values also showed consistent importance, emphasizing the influence of initialization on model outcomes. This finding aligns with the potential of ensemble learning to improve robustness and accuracy in hydrological predictions (Hosseini et al., 2025).

The Gini gains revealed variability in feature importance depending on geographic and climatic contexts. For example, Broadleaf Forest cover (BLF) and the coefficient of variation of flow were more influential in regions with variable flow regimes or significant forest contributions, affecting metrics like Alpha-NSE and Beta-NSE. Or, attributes like max slope and max temperature exhibit greater importance for metrics such as Missed-Peaks and FHV in catchments with distinct topographical and climatic characteristics, suggesting localized factors that affect hydrological responses.

The RF analysis highlights critical catchment attributes and hyperparameters that influence the performance of LSTM models in streamflow predictions. By effectively capturing nonlinear relationships and complex interactions, the RF model offers insights into the underlying drivers of model accuracy. Key takeaways include, 1) the significant role of hydrological attributes, such as streamflow and precipitation patterns, in enhancing model performance; 2) the influence of

505 land cover types and topographical features on predictive accuracy; and 3) the importance of
506 hyperparameter configurations, including input sequence length and seed values, in optimizing
507 LSTM outcomes.

508

509 *Figure 5. Feature importance*

510 *ranking for 15 top features for*

511 *different target metrics*

512 *derived from the trained*

513 *Random Forest model. This*

514 *figure highlights the relative*

515 *importance of various*

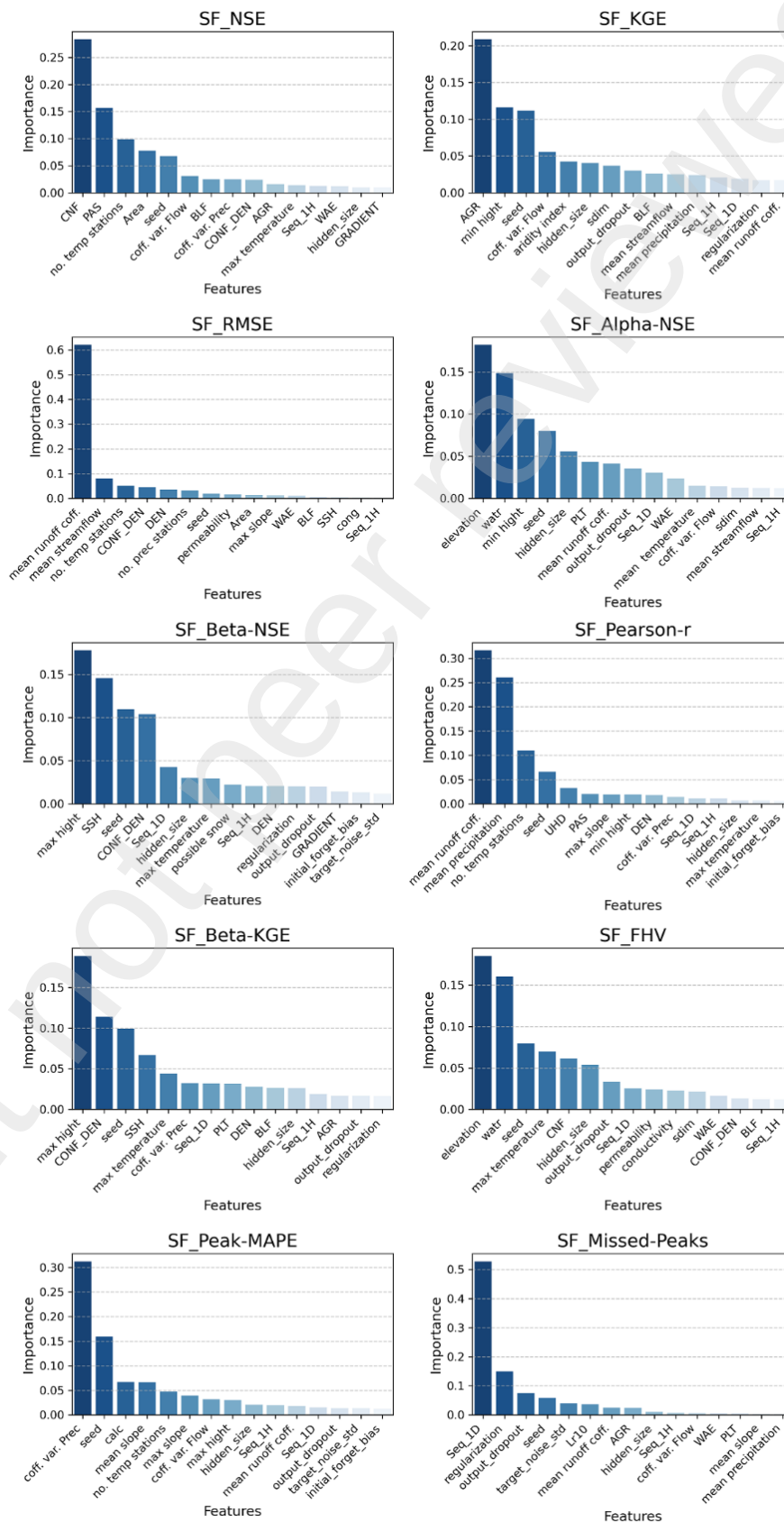
516 *features in predicting the test*

517 *metrics, emphasizing the*

518 *most influential attributes*

519 *shaping DNNs' accuracy.*

520



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3.2.2. Explainable AI Analysis through SHAP Summary Plots

The SHAP (SHapley Additive exPlanations) summary plots offer a comprehensive analysis of the 20 most influential features for different metrics—including catchment attributes and LSTM hyperparameters—shaping the performance of optimized regional LSTM models across multiple catchments in the Basque Country's humid, flashy hydrological regime. By examining 10 different hydrological performance metrics (NSE, KGE, RMSE, Alpha-NSE, Beta-NSE, Pearson-r, Beta-KGE, FHV, Peak-MAPE, and Missed-Peaks), the analysis elucidates both global and nuanced aspects of LSTM networks' behavior under varying hydrological conditions.

Influence of Catchment Attributes on LSTM Performance

The SHAP analysis reveals that the below catchment-specific hydrological and geomorphological characteristics are consistently dominant drivers of model performance across nearly all evaluated metrics:

Yearly mean runoff coefficient, streamflow, and precipitation emerge as the most impactful features, recurrently ranking at the top across NSE, KGE, RMSE, and Pearson-r. Their prominence reflects strong control over discharge dynamics and the data-driven model's capacity to capture basin-specific flow patterns. This aligns with classical hydrological understanding, where catchments with higher runoff generation and precipitation exhibit more predictable flow responses, favoring LSTM learning.

Catchment area and confluence density (CONF_DEN) significantly affect volumetric performance metrics such as NSE and RMSE. Larger catchments with denser drainage networks introduce spatial heterogeneity and complex routing effects, which challenge LSTM's ability to generalize learned patterns.

Slope-related features, including mean slope, maximum slope, and gradient, play a critical role in metrics related to extremes, noticeably for Peak-MAPE, FHV, and Missed-Peaks. These attributes reflect basin steepness, indicative of fast runoff generation and short response times, requiring the LSTM to learn rapid temporal dynamics effectively.

Land cover variables (e.g., pasture (PAS), agriculture (AGR), broadleaf forest (BLF), coniferous forest (CNF), shrubland (SSH), water surfaces (watr), wetlands (PLT), and denuded lands (DEN)) also exhibit substantial influence, particularly on KGE, NSE, and FHV. Their role underscores the effect of land use and vegetation on hydrological processes such as infiltration, evapotranspiration, and storage, which modulate streamflow generation and nonlinear responses to precipitation.

These findings provide empirical, data-driven validation of known hydrological principles, while also highlighting the capability of optimized regional LSTM networks to implicitly learn catchment-specific behaviors without direct access to these attributes during training.

Impact of LSTM Hyperparameters on Model Performance

In addition to catchment attributes, SHAP plots underscore the crucial influence of LSTM hyperparameters on regional model performance:

Sequence lengths daily and hourly (Seq_1D, Seq_1H) is consistently among the top influential hyperparameters across multiple metrics, with highest impact on Missed_peaks. Longer sequence lengths enable the model to capture longer temporal dependencies crucial for hydrograph reproduction; however, this could increase the computational costs.

Hidden size and learning rates are pivotal in shaping the model's capacity to learn complex flow patterns. Larger architectures generally enhance performance in metrics like KGE and FHV but may simultaneously increase vulnerability to errors in extreme flow metrics (e.g., NSE, Peak-MAPE), revealing a trade-off between model complexity and stability.

Dropout rate emerges as a critical regularization mechanism, influencing consistency and robustness across catchments. Properly tuned dropout rates reduce overfitting and enhance generalization, particularly noticeable in NSE, FHV, and KGE.

Batch size and loss function selections demonstrate metric-specific impacts, noticeably on NSE and Missed_peaks, suggesting their role in stabilizing model convergence and handling the variance in streamflow patterns. Larger batch sizes appear to smooth learning but may overlook localized peak behaviors.

Metric-Specific Findings and Hydrological Interpretations

Each performance metric yields distinct insights into how both catchment features and hyperparameters govern LSTM effectiveness:

NSE and KGE (global metrics) are primarily controlled by hydrological attributes—precipitation, area, runoff coefficient, and aridity index—highlighting their central role in shaping overall discharge reproduction. LSTM hyperparameters, particularly input sequence length and hidden size, are secondary but crucial to achieving high predictive skill.

RMSE, being sensitive to large deviations, is heavily influenced by slope, permeability, and confluence density, reflecting the difficulty in predicting extreme flows in steep or highly connected catchments.

Alpha-NSE and Beta-NSE, focusing on high and low flows respectively, show differential sensitivity to both catchment and LSTM architectural parameters, reflecting the need for targeted model adjustments to handle flow extremes.

FHV and Peak-MAPE, representing high-flow and peak-specific errors, are predominantly controlled by steepness and precipitation extremes, underscoring the need for models that can handle rapid responses and flow surges.

Missed-Peaks, a direct indicator of LSTM capacity to capture critical events, shows notable sensitivity to hyperparameters such as sequence lengths (Seq_1D, Seq_1H), regularization, dropout, hidden size, learning rate, batch size, and initial forget gate bias, demonstrating that architectural design strongly influences extreme event detection capability.

All in all, the ensemble of SHAP plots (Figure 6) offers a unified framework for understanding the interplay between catchment-specific hydrological behavior and LSTM model design. The results convey that hydrological features dominate overall model predictability, while hyperparameter tuning remains essential to optimize performance, particularly for flow extremes and peak events. Noticeably, the fact that LSTMs—trained solely on lumped hydrometeorological timeseries (precipitation, temperature, PET)—implicitly exhibit sensitivity to catchment attributes (not directly provided during training) illustrates the emergence of latent hydrological knowledge within the deep learning framework.

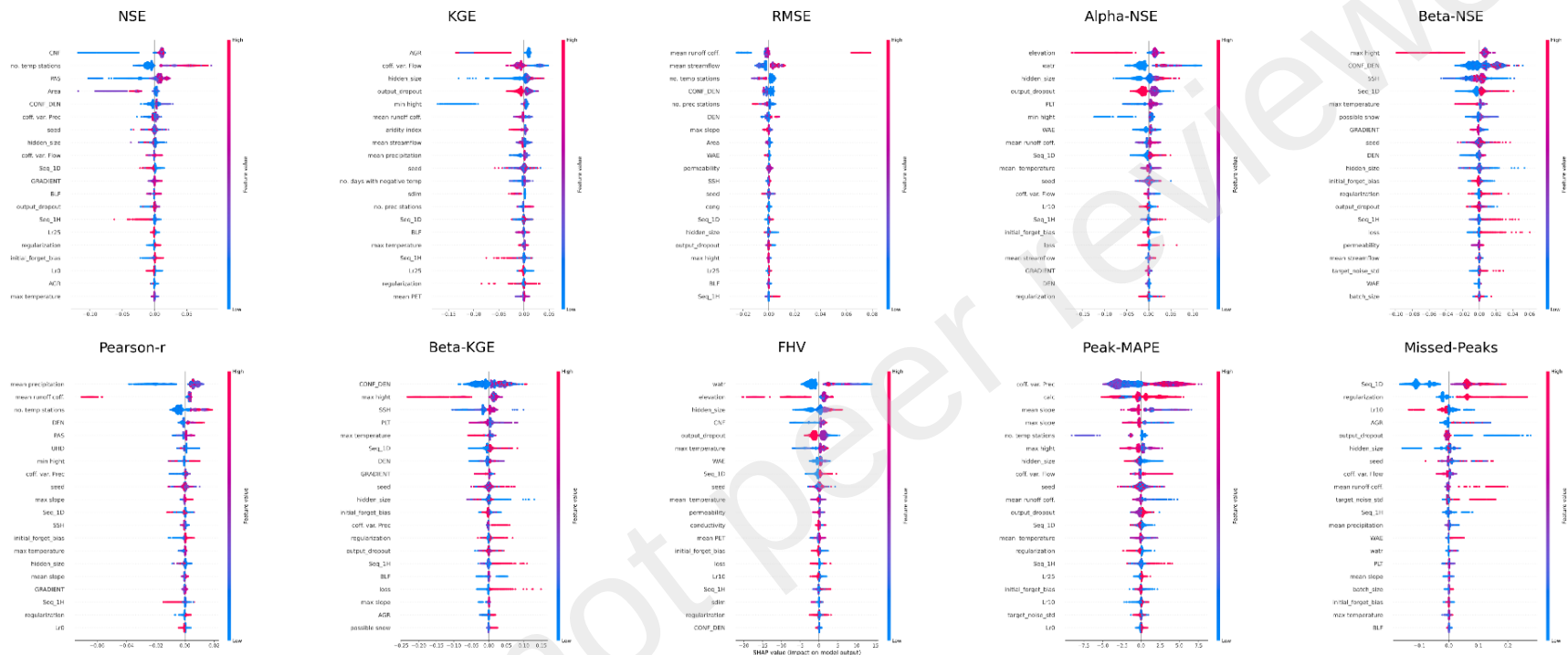


Figure 6. SHAP summary plots for the 20 most influential catchment attributes and LSTM hyperparameters affecting regional model performance across Basque Country catchments.

Each dot represents a single model-catchment instance, with color gradients showing feature magnitudes (red: high, blue: low). The SHAP value axis indicates feature positive and negative contribution to test performance metrics. Aggregated insights highlight dominant hydrological and architectural drivers of LSTM performance across diverse evaluation criteria.

3.3. Principal Component Analysis (PCA)

A Principal Component Analysis (PCA) was conducted on the test_DATASET, incorporating catchment attributes alongside the average NSE and KGE test performance metrics derived from optimized LSTM networks for streamflow across different catchments.

Figures 7 and 8 illustrate the PCA model results, including the scree plot (Figure 7) and the biplot analysis (Figure 8). The scree plot demonstrates that the first ten principal components cumulatively explain around 87.8% of the test_DATASET's total variance with three first PCs having near 60%. These 10 components capture the majority of the test_DATASET's variability, thus representing key aspects of the underlying structure. The biplot in Figure 8 visualizes the relationships between the first two principal components and the original features, indicating which catchment attributes contribute most significantly to each component. This figure shows a biplot from the local PCA analysis, illustrating how original catchment features contribute to the principal components and interact with one another. It visually represents the contributions of individual features to the first few principal components and highlights their relationships in the context of catchment hydrology.

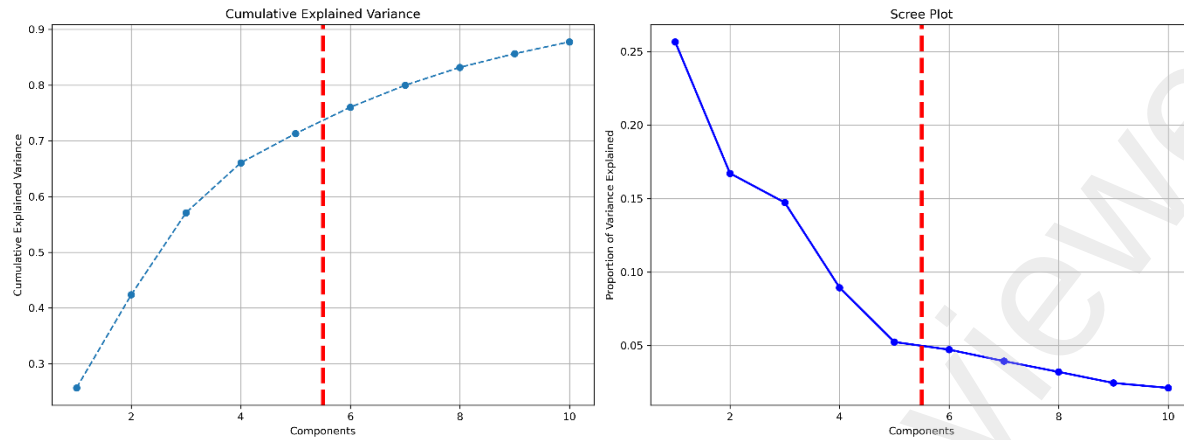


Figure 7. PCA results applied to the test_DATASET, showing the distribution of principal components. This plot illustrates how much variability is captured by each component, providing insights into the test_DATASET's structure.

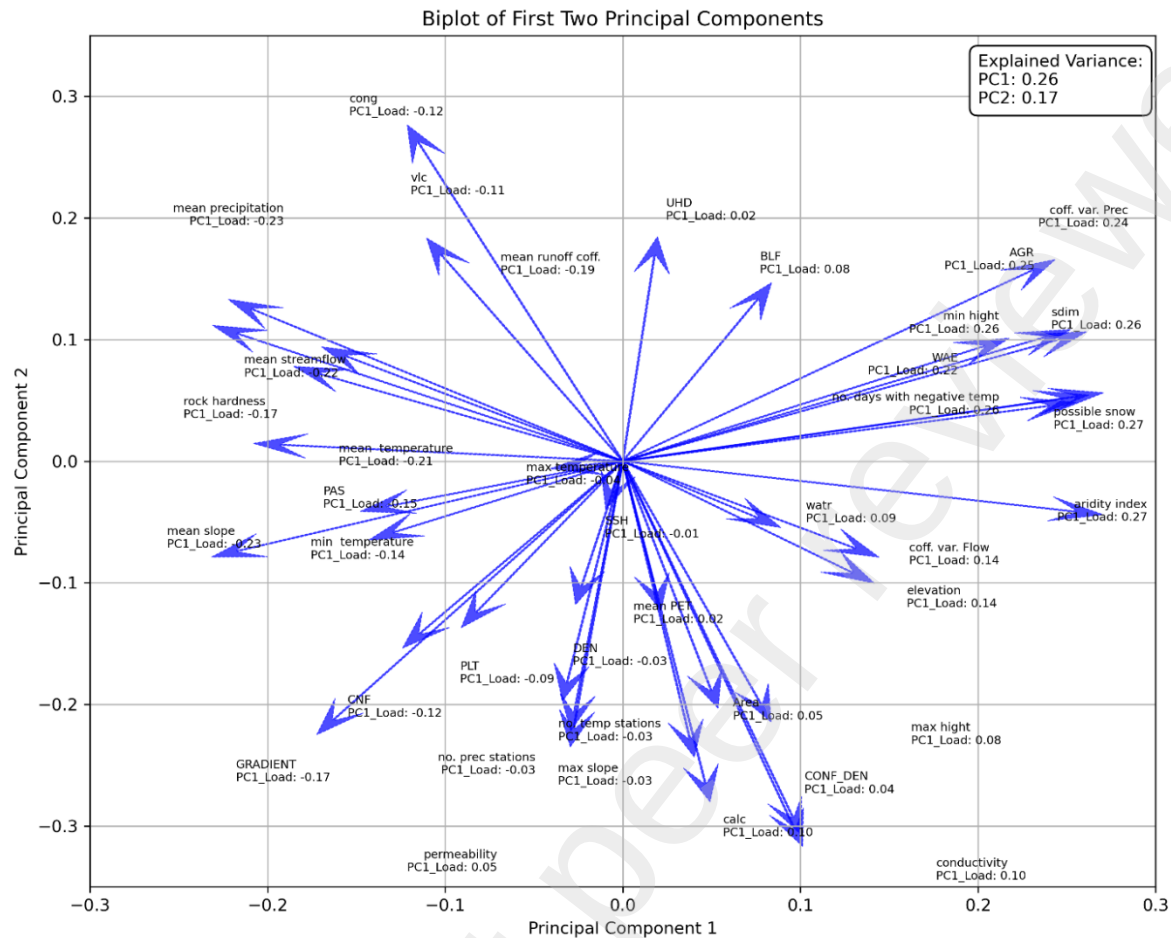


Figure 8. Biplot of the PCA analysis. This figure displays the principal components in relation to the original features, visually representing how each feature contributes to the principal components and their interactions.

Explained Variance and Component Loadings

Table 5 displays the component loadings and explained variance ratios, illustrating each principal component's contribution to the total variance. These loadings help identify key catchment features that influence the hydrological performance metrics of the optimized regional LSTMs.

Table 5. PCA components' loads and the explained variance ratios

Attributes	Principal components									
	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
Area	0.054	-0.203	-0.099	0.355	-0.247	-0.061	-0.053	-0.024	0.066	-0.036
CONF_DEN	0.040	-0.243	-0.068	0.344	-0.157	-0.054	-0.019	-0.127	0.058	-0.100
GRADIENT	-0.172	-0.225	-0.096	-0.106	-0.011	-0.018	-0.048	0.084	0.255	0.012
max slope	-0.030	-0.235	-0.108	0.118	0.188	0.247	-0.020	0.046	0.437	-0.144
mean slope	-0.232	-0.079	-0.198	-0.140	-0.094	-0.079	-0.137	0.036	0.006	0.007
elevation	0.142	-0.100	-0.295	-0.112	0.110	0.169	0.008	0.109	0.064	-0.037
min hight	0.255	0.048	-0.133	-0.126	0.119	0.107	0.117	-0.027	-0.128	-0.026
max hight	0.083	-0.214	-0.285	0.016	0.084	0.157	-0.011	0.092	0.111	0.057
UHD	0.019	0.185	0.248	0.158	-0.119	0.190	0.076	-0.052	-0.050	0.279
AGR	0.252	0.108	-0.053	0.008	-0.082	-0.120	0.101	0.143	-0.005	0.026
PAS	-0.148	-0.041	0.222	-0.121	-0.214	0.191	-0.056	0.088	0.153	-0.151
BLF	0.084	0.147	-0.229	0.091	0.067	0.306	-0.086	0.318	-0.117	-0.233
CNF	-0.124	-0.154	0.053	-0.247	-0.222	-0.262	0.056	-0.297	0.122	0.117
PLT	-0.091	-0.137	-0.111	0.036	0.278	-0.148	-0.272	-0.103	-0.288	0.324
SSH	-0.010	-0.039	0.151	0.298	0.390	-0.124	0.065	-0.119	0.060	-0.094
WAE	0.218	0.101	-0.090	0.044	-0.091	-0.248	-0.176	0.107	0.069	0.258
DEN	-0.027	-0.118	-0.068	0.032	0.337	0.028	0.451	-0.326	0.110	0.153
calc	0.100	-0.307	0.123	-0.178	0.068	-0.057	0.037	0.101	-0.131	-0.016
cong	-0.121	0.276	-0.102	0.145	-0.139	0.145	-0.095	-0.118	0.149	0.079
sdim	0.261	0.106	-0.050	0.035	-0.031	-0.093	0.056	0.067	-0.023	0.123
vlc	-0.110	0.183	-0.094	0.159	0.241	-0.237	0.176	-0.001	0.009	-0.276
watr	0.089	-0.055	-0.118	0.250	-0.061	0.028	-0.378	-0.389	-0.130	-0.114
conductivity	0.101	-0.317	0.128	-0.153	0.031	0.077	0.023	0.111	-0.017	0.079
permeability	0.049	-0.280	-0.114	-0.175	0.044	-0.113	-0.123	0.183	-0.292	0.031
rock hardness	-0.170	0.094	-0.228	0.006	0.150	-0.244	-0.124	-0.015	-0.166	-0.322
no. prec stations	-0.029	-0.222	-0.042	0.279	-0.147	-0.034	0.172	0.203	-0.141	0.099
no. temp stations	-0.034	-0.197	-0.078	0.326	-0.220	-0.094	0.224	0.090	-0.121	0.098
possible snow	0.271	0.056	-0.166	-0.002	-0.071	-0.075	0.021	-0.090	-0.006	0.019
no. days with negative temp	0.262	0.054	-0.186	-0.009	-0.072	-0.093	0.024	-0.052	0.010	0.078
mean runoff coeff.	-0.185	0.078	-0.116	0.078	0.069	0.207	0.321	0.112	-0.211	0.292
aridity index	0.270	-0.044	0.143	0.124	0.036	0.098	-0.037	-0.009	0.042	-0.046
mean precipitation	-0.231	0.112	-0.218	-0.094	-0.053	-0.044	-0.046	-0.069	-0.036	0.025
mean streamflow	-0.222	0.133	-0.222	-0.001	0.015	0.051	0.112	0.017	-0.117	0.152
mean temperature	-0.208	0.015	0.232	0.147	-0.011	0.014	-0.126	0.162	-0.131	0.050
min temperature	-0.143	-0.065	0.187	0.163	0.070	0.080	-0.076	0.206	-0.270	-0.126
max temperature	-0.041	-0.006	0.114	0.127	0.366	-0.066	-0.350	0.142	0.250	0.386
coeff. var. Prec	0.243	0.166	0.136	-0.007	0.119	0.050	-0.151	0.040	0.013	0.051
coeff. var. Flow	0.144	-0.079	0.263	-0.054	0.105	0.127	0.117	-0.120	-0.208	-0.242
mean PET	0.021	-0.123	-0.027	-0.059	-0.006	0.483	-0.140	-0.425	-0.290	0.045
Explained Variance Ratio	25.7%	16.7%	14.7%	8.9%	5.2%	4.7%	3.9%	3.2%	2.5%	2.1%
Cumulative Variance	25.7%	42.4%	57.1%	66.1%	71.3%	76.0%	80.0%	83.2%	85.6%	87.8%

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Explained Variance: The first principal component (PC1) captures approximately 25.7% of the variance, with PC2 and PC3 accounting for 16.7% and 14.7%, respectively. The first five components collectively account for 71.3% of the variance, while the first ten components cover around 87.8%, providing a comprehensive view of the test_DATASET.

Component Loadings: According to Table 5, the PCA loadings reveal each feature's influence on the principal components:

PC1: Climatic and topographic attributes dominate, explaining 23.9% of the variance. Features with high positive loadings include the probability of snowfall (0.270), aridity index (0.270), area occupied by sedimentary soils (0.261), elevation (0.255), and agricultural land area (0.252). Meanwhile, attributes such as average precipitation (-0.230), mean runoff coefficient (-0.184), average temperature (-0.208), and average slope (-0.231) show significant negative loadings. This pattern suggests that hydrological dynamics are strongly influenced by both climatic (e.g., aridity, precipitation) and topographical factors (e.g., elevation, slope), highlighting complex hydrological responses in relation to these variables. High aridity and elevation values correspond to increased runoff variability, while higher precipitation and temperature appear to dampen streamflow variability.

PC2: Geological and morphological characteristics are predominant in PC2, capturing 16.5% of the variance. Features with high positive loadings include river confluence density (CONF_DEN: 0.284), average soil conductivity (0.267), area of calcareous rocks (0.257), catchment size (0.249), and max slope (0.232). Conversely, the area occupied by conglomerate rocks (-0.232) has a strong negative loading. These loadings suggest that the density of river confluences and soil type diversity are critical factors shaping catchment hydrological behavior, with more complex confluence networks likely introducing additional variability in hydrological responses.

PC3: Vegetation cover and land use attributes are influential in PC3, explaining 13.8% of the variance. High loadings include maximum (0.258) and average elevation (0.276), area covered by

broadleaf forests (0.247), and bedrock hardness (0.236). Average precipitation (0.225) shows a positive loading, while coefficient of flow variation (-0.274) and pasture lands (-0.229) and urban area (-0.214) have negative loadings. This component suggests that broadleaf forests enhance water retention, while urbanization may increase impervious surfaces, complicating runoff dynamics. Negative loadings for flow variability and urban areas reflect the crucial role of land cover in hydrological behavior.

Other Components: Attributes such as soil conductivity, permeability, and anthropogenic influences further contribute to variance in the higher components. These factors highlight the impact of land use and soil properties on natural hydrological processes, modifying water storage and flow patterns.

Interpretation of PCA Outcomes

The PCA results reveal the interconnectedness of catchment attributes in driving hydrological behavior:

Component 1 (PC1): Highlights the combined influence of climatic and topographical factors, suggesting that aridity and elevation enhance runoff variability while precipitation and temperature have a buffering effect that facilitate predictions by LSTM networks.

Component 2 (PC2): Emphasizes the importance of catchment morphology and geology. Catchments with higher confluence densities and unique soil types exhibit increased variability in hydrological responses, reflecting the complex geological landscape's impact that makes it harder for LSTMs to accurately predict runoff from rainfall events.

Component 3 (PC3): Underlines the significance of vegetation cover and land use. Broadleaf

forests help retain water, while urban areas disrupt natural runoff processes. Flow variability is a

distinguishing factor, with specific land cover types amplifying or mitigating runoff responses.

These behaviors can indirectly affect LSTMs' performance in different locations and catchments.

The PCA outcomes underscore the intricate nature of hydrological dynamics in catchments,

where climate, topography, geology, and land use interact in complex ways to shape runoff and

streamflow behavior. By interpreting optimized LSTM performance through PCA, we gain insights

into these interrelationships, which can guide future modeling efforts and improve rainfall-runoff

predictions under changing environmental conditions. These findings are especially valuable for

enhancing regional hydrological deep learning (DL) models, like DNN LSTMs, which are critical for

managing water resources amid variable climatic scenarios. The results highlight the potential of

PCA-informed approaches to optimize DNNs in hydrology, enabling more accurate rainfall-runoff

predictions in response to complex, shifting climate patterns.

4. Discussion

4.1. Regional Hydrological Artificial Intelligence

Although the optimized LSTM networks did not directly incorporate catchment attributes—

such as soil type, land cover, or elevation— while being trained, the observed correlation

between model performance and catchment characteristics would suggest that the optimized

LSTMs effectively learned implicit representations of these unique attributes. In other words,

they learned how to discover the fingerprints of every catchment from during a regional training

on only on the hydrometeorological data. This hidden knowledge was likely derived from the input variables and their well-optimized hyperparameters.

Key Findings and Implications from correlation analyses: This analysis underscores that catchments with stable hydrological regimes—characterized by high runoff coefficients, steady precipitation, and certain vegetation types like coniferous forests—are more conducive to accurate predictions by LSTM networks. In contrast, catchments with high climate variability, extreme flow fluctuations, complex geological features, or significant human modifications (e.g., agriculture, controlled flows) challenge DNN model accuracy.

Despite the absence of explicit catchment attributes in the training phase, the optimized regional LSTM models appear to have implicitly learned latent hydrological patterns and what we know from physical hydrology. These patterns likely reflect underlying processes and environmental interactions unique to specific catchments. Overall, this correlation analysis offers critical insights for refining DNNs, guiding future optimization efforts, and improving the understanding of how catchment characteristics influence hydrological predictions in DLs.

Intersection of physical hydrology and AI/DL models

The findings reveal the intricate relationships between catchment attributes and the performance metrics of optimized regional Long Short-Term Memory (LSTM) networks in rainfall-runoff modeling. Despite the absence of explicit catchment features in the training phase, the LSTM networks showcased a remarkable ability to capture complex latent relationships inherent in hydrometeorological data. This capability underscores the potential of deep learning (DL)

techniques and hyperparameter optimization to interpret underlying hydrometeorological dynamics, aligning with advances in machine learning that highlight deep neural networks' strengths in identifying patterns within extensive datasets, often exceeding the capabilities of traditional models.

Moreover, integrating both timeseries data and catchment attribute information into the training process could expedite model convergence and improve the fitting process, leading to computational efficiency gains. By incorporating both hydrometeorological and environmental features, DL models can uncover critical relationships earlier, facilitating more efficient training and enhancing predictive accuracy. This advocates for a multifaceted approach to DL model training that incorporates diverse factors instead of relying solely on historical timeseries data.

Additionally, the LSTM networks demonstrated ability to discern relationships from extensive hydrometeorological datasets highlights their potential as powerful modeling tools in hydrology. These findings suggest that LSTMs are effective in capturing nonlinear relationships and temporal dependencies often present in hydrological systems, reinforcing the cautions presented by Kratzert et al. (2024) against training LSTMs exclusively on single catchments. Training on data from multiple catchments is essential for capturing the variety of hydrological behaviors across geographic and climatic contexts, helping to avoid overfitting to individual water basin characteristics (the catchments' uniqueness paradigm, Beven, 2020) and enhancing model robustness and generalizability.

The implications of these results extend beyond model performance to water resource management strategies. An improved understanding of catchment attributes and hydrological

responses aids informed decision-making in water resource management, particularly amidst the era of climate variability. As hydrological extremes become more frequent, accurate predictions are crucial for effective management and mitigation.

4.2. Interpreting the Hydrological Relevance of AI Explainability Insights

The combined interpretation of Random Forest (RF) Gini gains and SHAP summary plots consistently highlights that a combination of catchment-specific hydrological attributes and AI-related hyperparameters governs LSTM networks performance in regional rainfall-runoff modeling across the Basque Country's humid and flashy catchments. Together, these explainability approaches provide robust and complementary insights into the factors shaping DNN's skill, supporting both hydrological reasoning and AI-specific considerations.

Hydrological attributes related to catchment behavior and rainfall-runoff processes dominate as key factors influencing LSTM accuracy and robustness. Specifically, mean yearly streamflow, precipitation seasonality, precipitation intermittency (prec_cv), and aridity index emerge as top contributors to model skill in both Gini and SHAP analyses. These variables reflect essential hydrological controls: streamflow magnitude and variability act as proxies for catchment responsiveness, hydrological stability, and memory effects, while precipitation regime characteristics inform the timing and intensity of runoff responses. The positive SHAP effects of streamflow on accuracy metrics (e.g., NSE, KGE) suggest that LSTMs perform best where hydrological signals are stable and continuous, while in highly variable or intermittent systems, generalization becomes more challenging. Furthermore, the aridity index highlights model

sensitivity to intermediate climatic conditions, where a balance between predictability and hydrological complexity may govern LSTM learning capacity.

The role of land cover attributes, specifically coniferous forest (CNF), pasture (PAS), and agriculture (AGR), underscores the influence of vegetation as well as land use and human intervention on streamflow predictability. Coniferous forests generally stabilize flow regimes, dampening peaks and supporting sustained baseflows, which facilitates model learning. This is confirmed by their positive SHAP associations with model accuracy and negative associations with error metrics (e.g., Missed-Peaks, Bias). In contrast, pasture and agricultural areas, although important, are often associated with higher uncertainty and anthropogenic variability, as indicated by SHAP analyses showing negative influences on model accuracy in many catchments. This suggests that land use heterogeneity and human influences introduce additional complexity that challenges LSTM generalization.

Wetlands (WAE) and water bodies (watr) also exert important influences, pointing to eco-hydrological processes such as time-lagged responses, storage effects, and regulated discharges. These factors introduce delayed and nonlinear dynamics that are difficult for LSTMs to capture, explaining why their presence is often associated with reduced accuracy and higher Missed-Peaks as shown in SHAP summary plots.

Topographical features, especially maximum slope, emerge as important in the context of extreme flow events. Steep catchments, prone to quick runoff and flash floods, require LSTM models to learn fast-response dynamics. SHAP results show that higher slopes are associated

with better peak capture (lower Missed-Peaks) but sometimes at the cost of increased bias, reflecting the trade-off between representing extremes and maintaining overall balance.

Hyperparameter Optimization: Hydrologically Informed AI Design

An important and novel insight emerging from both RF Gini gains and SHAP analyses is the critical role of LSTM hyperparameters in shaping model performance, underscoring the need for hydrologically informed AI design. Among these, input sequence length—which determines how much antecedent hydrological memory is provided—shows one of the highest SHAP impacts across all performance metrics. The consistent positive influence of sequence length on accuracy and error reduction confirms that longer memory windows are beneficial for capturing delayed runoff processes, especially in flashy catchments with short response times. However, SHAP analyses also reveal diminishing returns and even negative impacts for excessively long sequences, signaling potential overfitting or dilution of relevant information. This highlights a critical trade-off between capturing hydrological memory and maintaining model generalization.

Other LSTM architectural hyperparameters, including hidden size and dropout rate, also exhibit substantial influence. Hidden size, which controls model capacity, must be carefully balanced to capture flow variability without overfitting, as evidenced by SHAP's mixed but significant contributions. Similarly, dropout rates, essential for regularization, show dual roles—improving generalization but, when excessively high, hampering the model's ability to learn fine-grained patterns, as revealed in both accuracy and error-based SHAP plots.

Interestingly, random seed (seed), which governs the stochastic initialization of LSTM weights, appears frequently among influential factors in both RF and SHAP results. This

underlines the sensitivity of LSTM networks to initialization, a key challenge in deep learning (Sutskever et al., 2013; Glorot and Bengio, 2010). The results support ensemble modeling approaches, where training multiple models with different seeds can reduce variance and improve robustness (Hosseini et al., 2025).

Altogether, this joint SHAP and Gini analysis provides a comprehensive and hydrologically coherent narrative:

- Catchment hydrological attributes, especially those related to flow magnitude, precipitation regime, and aridity, fundamentally control LSTM skill and generalization.
- Land cover and topography modulate flow predictability and extremes, adding complexity that challenges AI models.
- LSTM hyperparameters, specifically input sequence length, hidden size, and dropout, need careful tuning guided by hydrological insights to ensure optimal performance.
- The sensitivity to random seed emphasizes the importance of ensemble learning to achieve stable and generalizable predictions for multi-objective tasks such as regional hydrological rainfall-runoff modeling in different catchments.

4.3. Future Research Directions

Expanding Hyperparameter Optimization Techniques: While random search proved effective in hyperparameter optimization, future studies could explore more sophisticated techniques, such as Bayesian optimization, to refine model tuning further. Incorporating methods like clustering-based optimization or alternative ensemble strategies may identify better configurations that balance accuracy with computational efficiency. Uncertainty

quantification methods (Klotz et al., 2022), including Bayesian LSTMs, Monte Carlo dropout, and Mixture Density Networks, could also enhance prediction reliability, an important consideration as climate variability introduces heightened risk in regions prone to hydrological extremes.

Hybrid Modeling Approaches: Integrating LSTM models with traditional physically-based hydrological models present an opportunity to leverage the strengths of both approaches. Hybrid models combining physically-based principles with data-driven insights could improve accuracy across multiple timescales and catchment types, extending the applicability of LSTMs to varied hydrological environments. Moreover, further exploration of DL ensemble models, particularly those that combine LSTMs with Transformers (Vaswani et al., 2017), could increase robustness, enabling better performance in regions affected by seasonal shifts and climate-driven hydrological changes.

Extending Model Validation Across Diverse Regions: To validate the robustness of optimized LSTM frameworks, future studies should apply these methods in regions with diverse hydrological and climatic characteristics. Evaluating model flexibility in different environments is essential as climate change amplifies hydrological extremes. Moreover, integrating DL models with real-time environmental data from remote sensing and Internet of Things (IoT) technologies could further enhance adaptability, enabling timely responses to climate-affected water resource needs.

Enhancing Interpretability with Explainable AI (xAI): Future research should prioritize improving the interpretability of DL models through more different and advanced Explainable AI (XAI) techniques, fostering collaboration between AI specialists and hydrologists. By clarifying

how DL models weigh different features and adjust to new conditions, xAI could increase trust in AI-driven hydrological forecasting, particularly in applications sensitive to climate-induced variability. Moreover, the integration of SHAP-based interpretability in this study enhances transparency and provides actionable insights into model design for operational hydrology. Future research should investigate dynamic DL architectures (e.g., attention mechanisms) and hybrid physics-informed neural networks to improve model adaptability and robustness in complex catchments.

5. Conclusion

This study has demonstrated the potential of hyperparameter-optimized regional Long Short-Term Memory (LSTM) networks to effectively learn rainfall-runoff dynamics across diverse hydrological catchments in the Basque Country, Spain. By training LSTM models solely on hydrometeorological inputs—precipitation, temperature, and potential evapotranspiration—without direct access to catchment-specific attributes, we showed that these models can implicitly capture latent hydrological features that govern streamflow behavior, achieving robust generalization across humid and flashy basins.

To better understand the factors shaping the regionally optimized LSTMs' predictive accuracy, we designed a comprehensive explainable AI (xAI) framework, combining correlation analysis, Random Forest (RF) modeling (Gini gains and SHAP summary plots), and Principal Component Analysis (PCA). These complementary methods consistently revealed that both catchment-specific hydrological attributes and LSTM hyperparameters

868 jointly control model skill. Noticeably, mean yearly runoff coefficient, streamflow
869 magnitude, and precipitation coefficient of variation (precipitation intermittency)
870 emerged as key determinants of LSTM accuracy, emphasizing that catchments with
871 clearer and more stable hydrological signals are easier for LSTMs to predict. In contrast,
872 steep, highly variable, and drier catchments posed greater challenges due to their
873 complex and nonlinear responses.

874 Furthermore, SHAP analyses provided new insights into model behavior, showing that
875 larger catchments and wetter basins with higher runoff coefficients systematically lead to
876 improved LSTM performance, while high precipitation intermittency and extreme slopes
877 degrade model skill by introducing complex dynamics. These results reinforce classical
878 hydrological understanding through data-driven AI evidence, confirming that LSTM
879 models are more effective in environments where hydrological processes are consistent
880 and pronounced.

881 Importantly, LSTM hyperparameters such as input sequence length, hidden size, and
882 dropout rate were consistently identified as critical to model success. The input sequence
883 length—capturing antecedent hydrological memory—had one of the highest impacts on
884 performance, confirming that longer memory windows help LSTMs better represent
885 delayed runoff responses in flashy catchments. However, the diminishing returns
886 observed beyond certain thresholds underline the need for careful balance to avoid
887 overfitting. Additionally, model complexity (hidden size) and regularization (dropout rate)

888 influence the trade-off between capturing hydrological variability and maintaining
889 generalization, emphasizing that systematic hyperparameter optimization is essential
890 when deploying LSTM models in complex regions.

891 From a methodological standpoint, this research makes a significant contribution to
892 bridging the gap between the black-box nature of AI and deep learning models and
893 hydrological interpretability. By combining optimized LSTM architectures with robust
894 explainability tools, we offer practical guidelines for both AI modelers and hydrologists,
895 underscoring the importance of integrating hydrological domain knowledge into AI
896 design. These findings support the development of more transparent, reliable, and
897 generalizable AI models for rainfall-runoff modeling.

898 Moreover, this study emphasizes that while regional LSTMs trained solely on
899 meteorological data can capture key hydrological dynamics, incorporating catchment
900 attributes as auxiliary inputs could further improve model accuracy, convergence speed,
901 and robustness, especially in data-scarce or highly variable regions.

902 Ultimately, the implications for water resource management and flood risk
903 assessment are profound, particularly in regions prone to flash floods like the Basque
904 Country. By improving streamflow predictions through optimized and explainable DL
905 models, this research contributes to advancing early warning systems, flood mitigation
906 strategies, and sustainable water management. As climate variability continues to affect
907 hydrological systems globally, AI-driven approaches like LSTM networks—enhanced

through explainability and optimization—represent a promising pathway toward more accurate and interpretable hydrological forecasting models. This work provides key insights into the generalization capacity of deep learning models in hydrology, advocating for continued exploration of hybrid approaches that blend AI methodologies with hydrological science, thus fostering trust and broader adoption of AI in environmental modeling.

6. Codes, Data and reproducibility

The dataset and all related codes utilized for this research, along with comprehensive instructions for replicating the experiments, are accessible on our repositories on <https://doi.org/10.5281/zenodo.15080569>, <https://doi.org/10.5281/zenodo.15080600>, https://github.com/farzadhoseini/xAI_LSTMs_BasqueCountry, <https://github.com/farzadhoseini/ensemble.deep.learning>, [https://github.com/farzadhoseini/Precise Tuning of Regional Hydrological LSTM Networks](https://github.com/farzadhoseini/Precise_Tuning_of_Regional_Hydrological_LSTM_Networks). We prioritize transparency and reproducibility so that fellow researchers and practitioners can verify our findings and employ the same codes for hyperparameter optimization, ensemble learning, and Explainable AI techniques of their research and applications.

7. Author contributions

Farzad in his PhD research, conceptualized the study, curated and analyzed data, conducted formal analyses, investigated findings, and crafted the original draft of the manuscript. His input was crucial in refining and editing the final version. Professor Dr. Cesar Álvarez Diaz and Dr.

Cristina Prieto Sierra provided valuable supervision, offering guidance in conceptualization, methodology, and ensuring the integrity of the work. Their contributions extended to reviewing and editing the manuscript, ensuring its coherence and accuracy.

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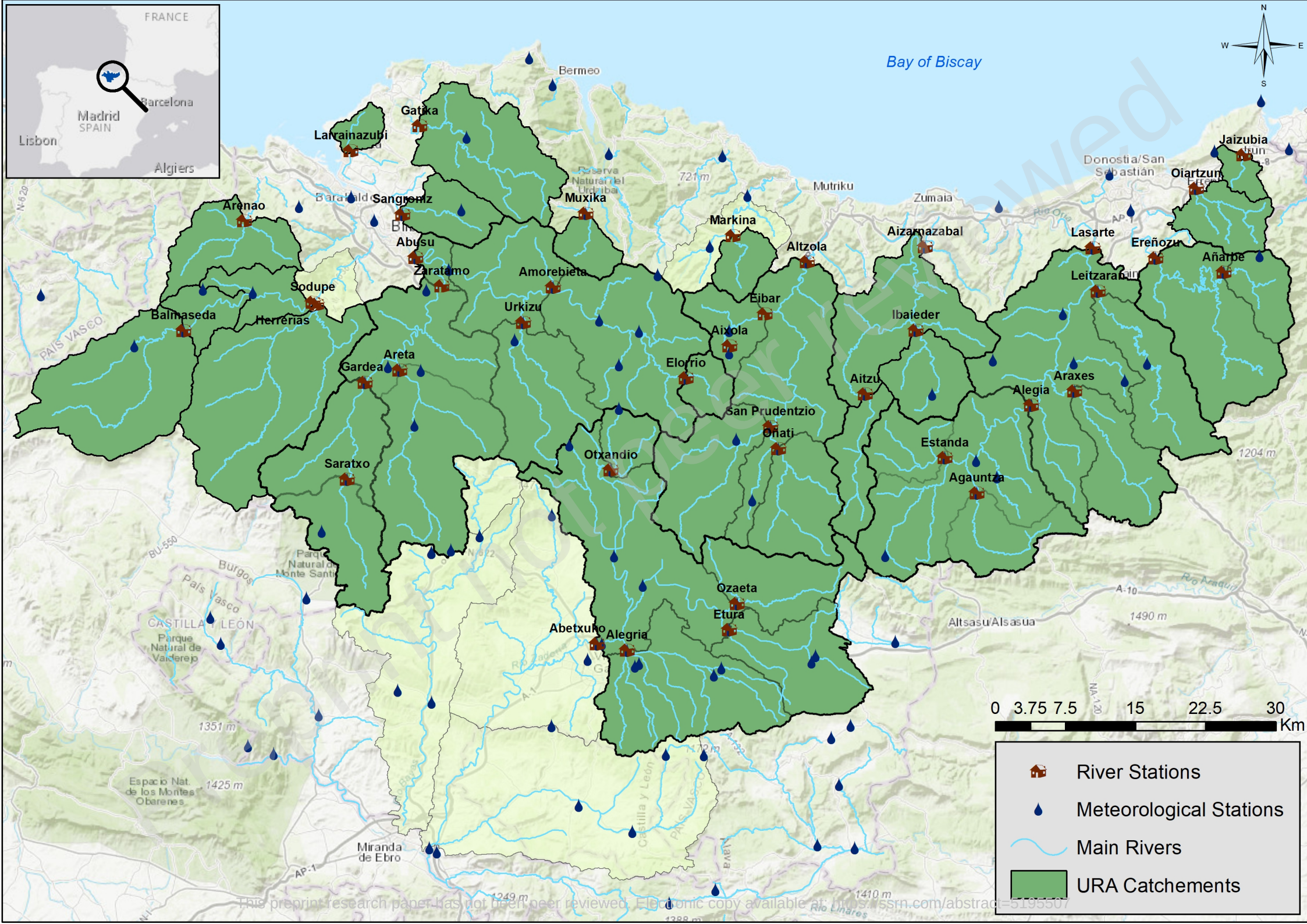
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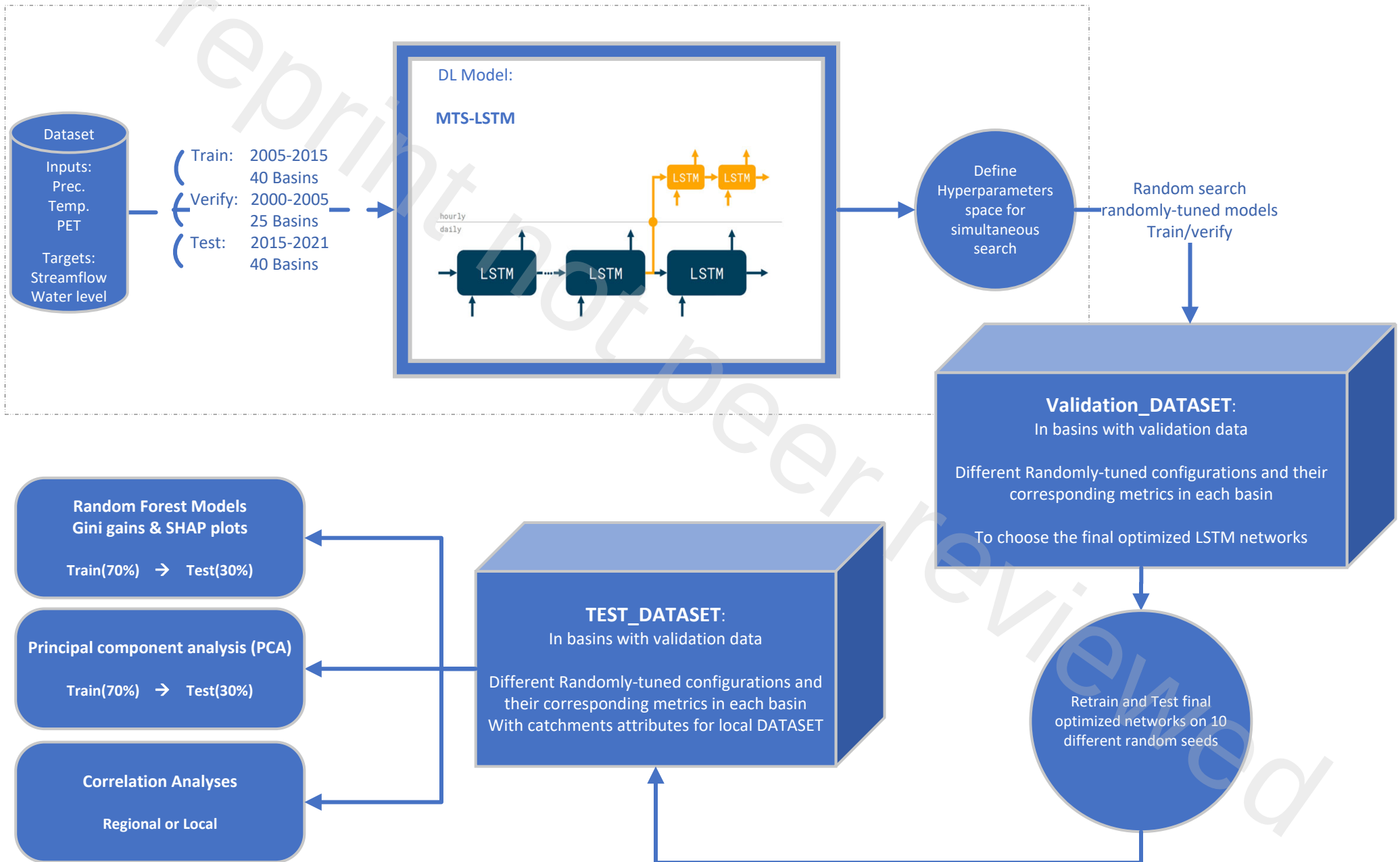
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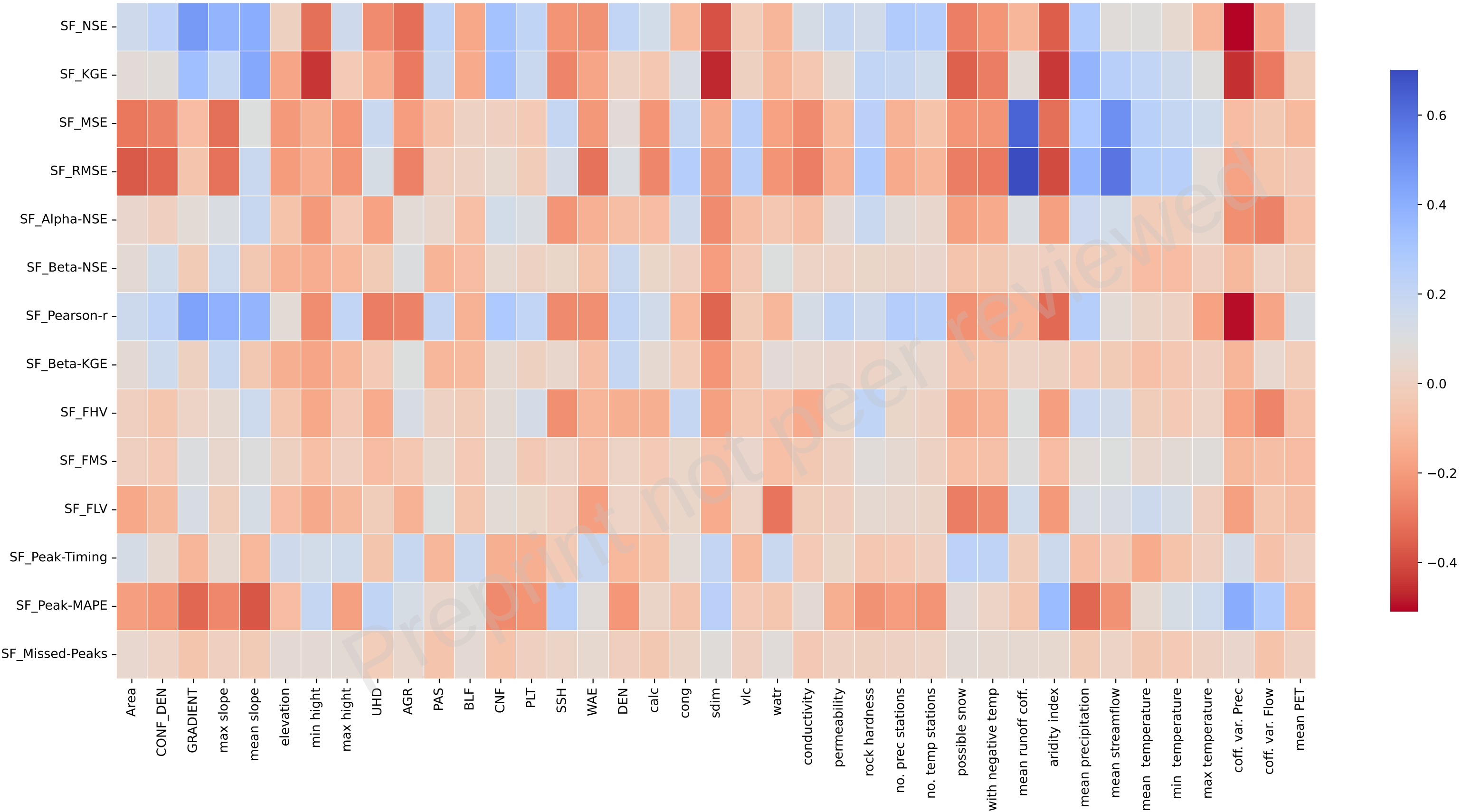
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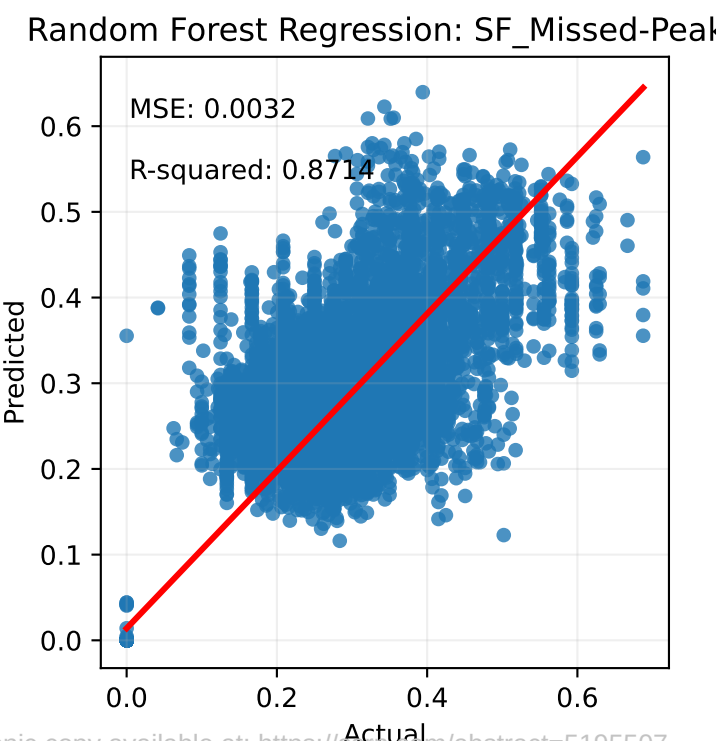
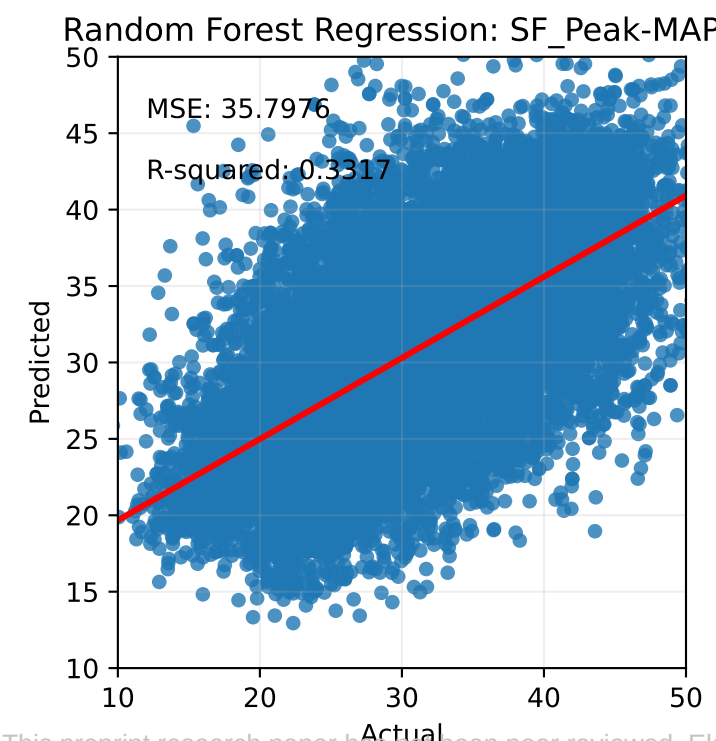
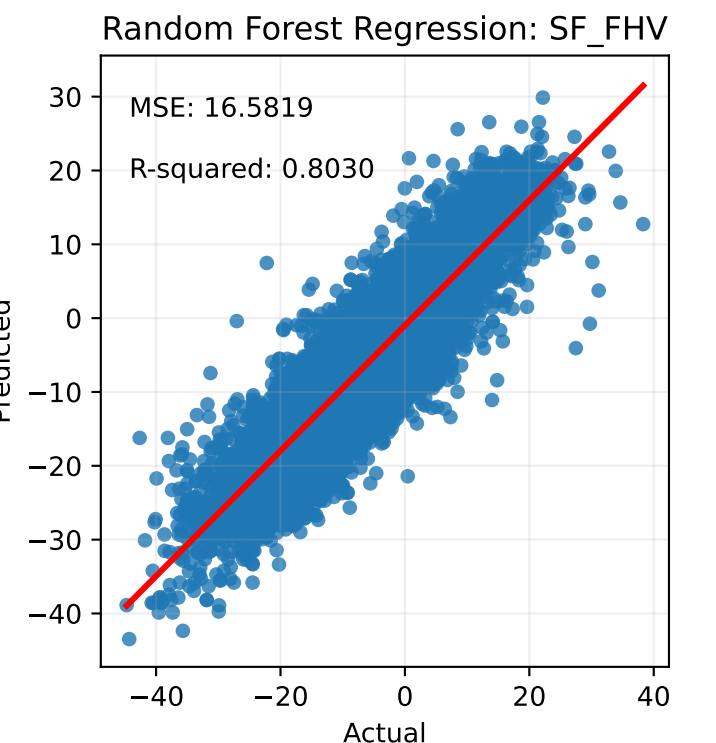
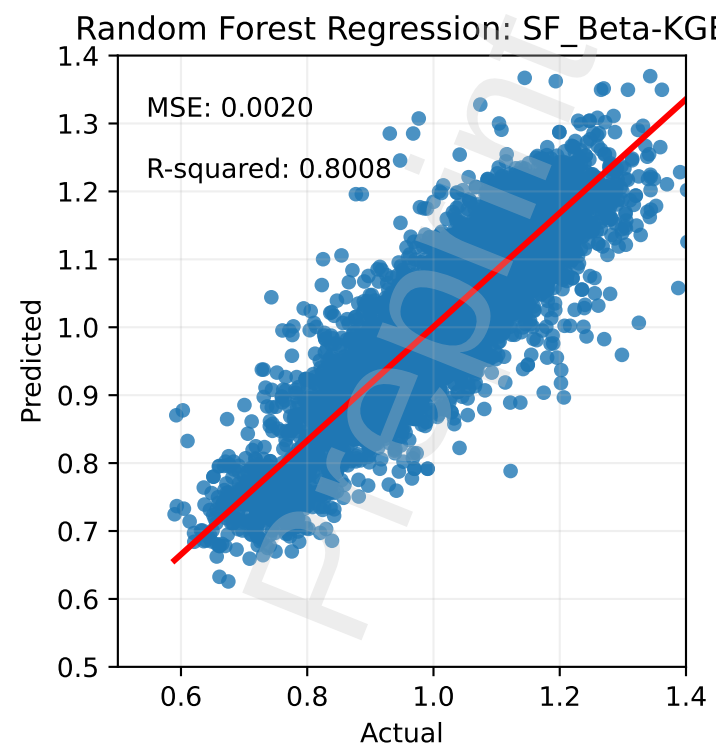
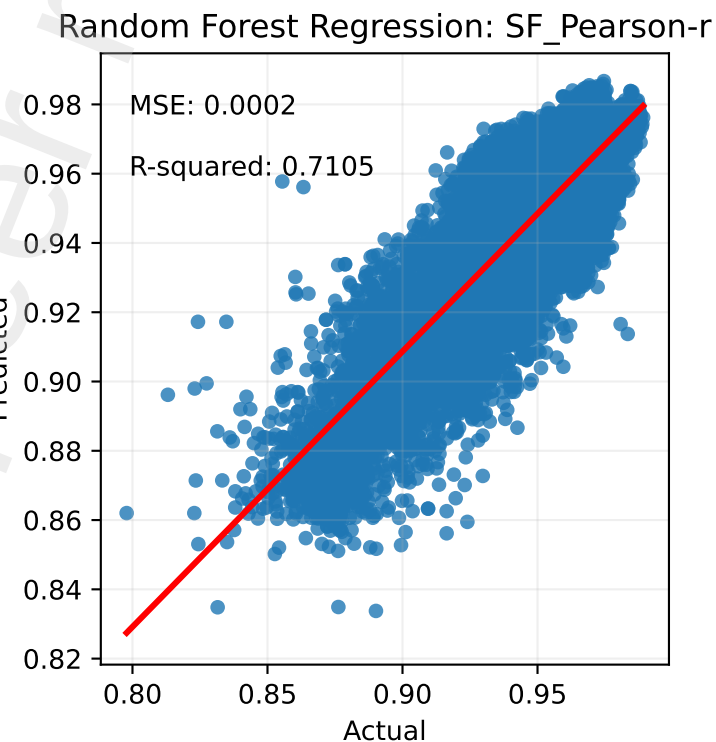
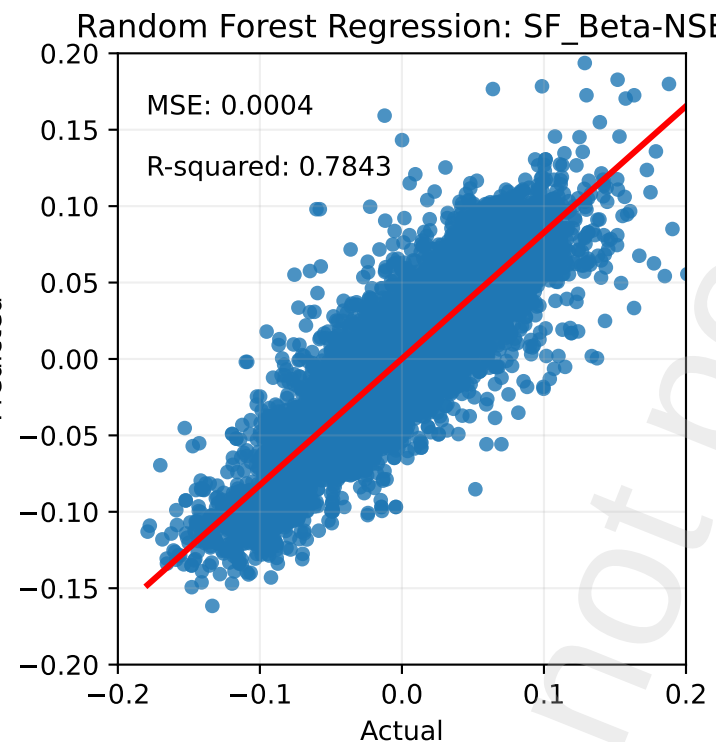
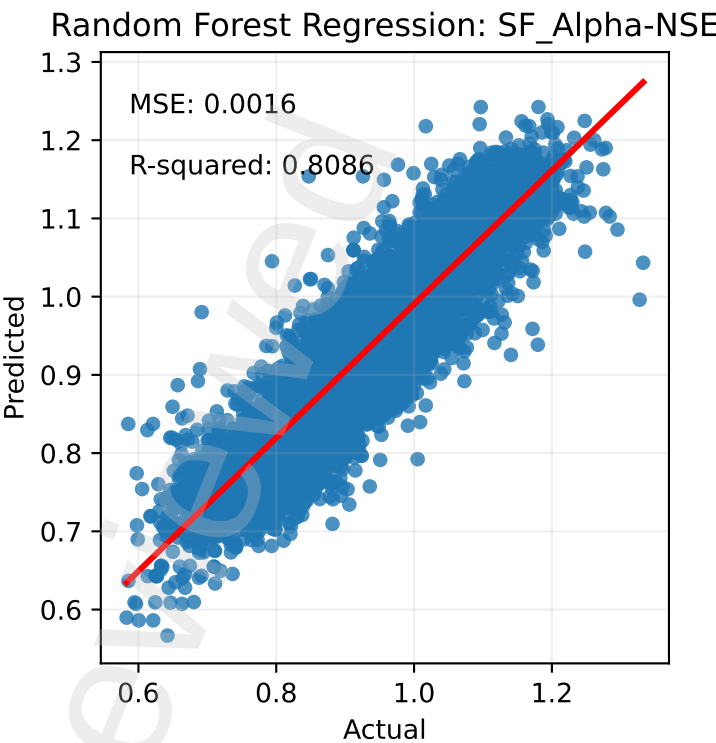
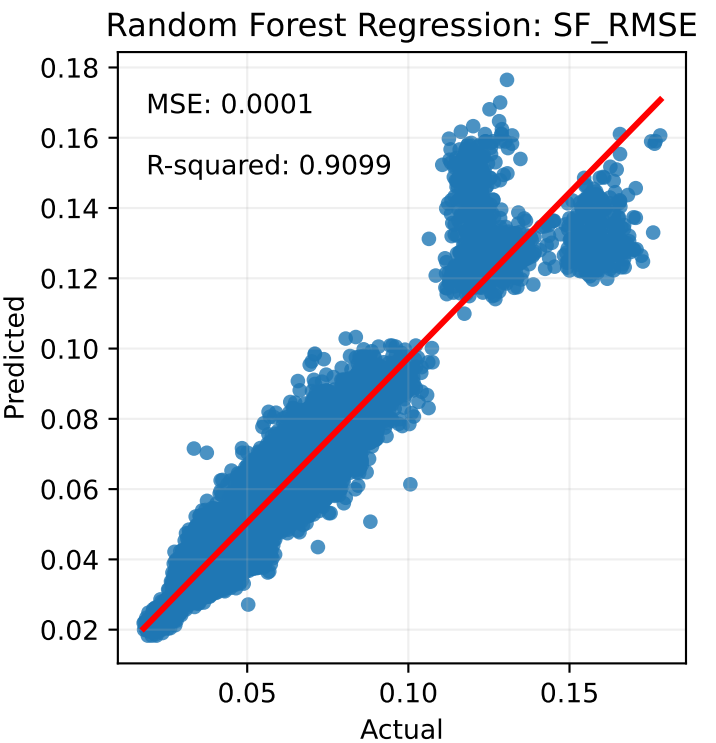
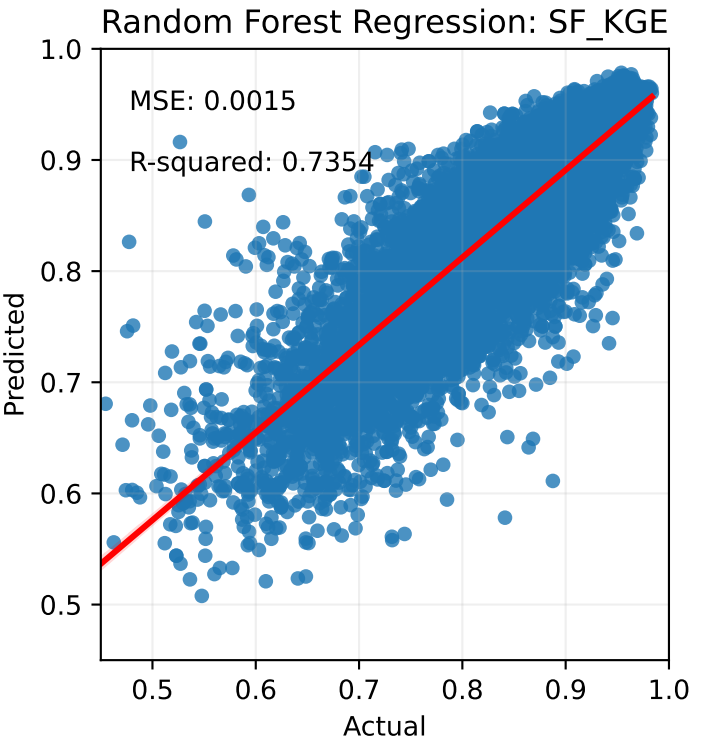
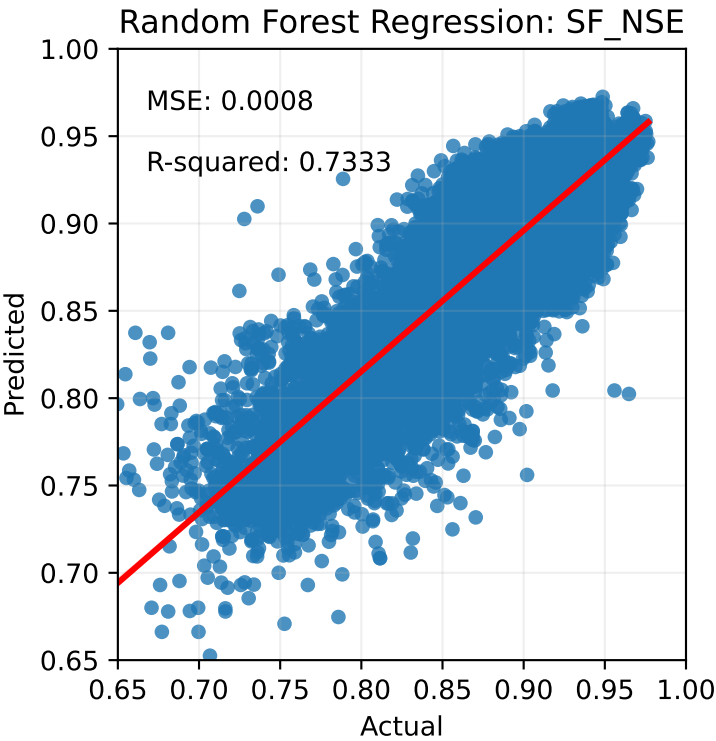


Methodology employed for assessment of importance of different catchments' attributes on LSTMs' performance

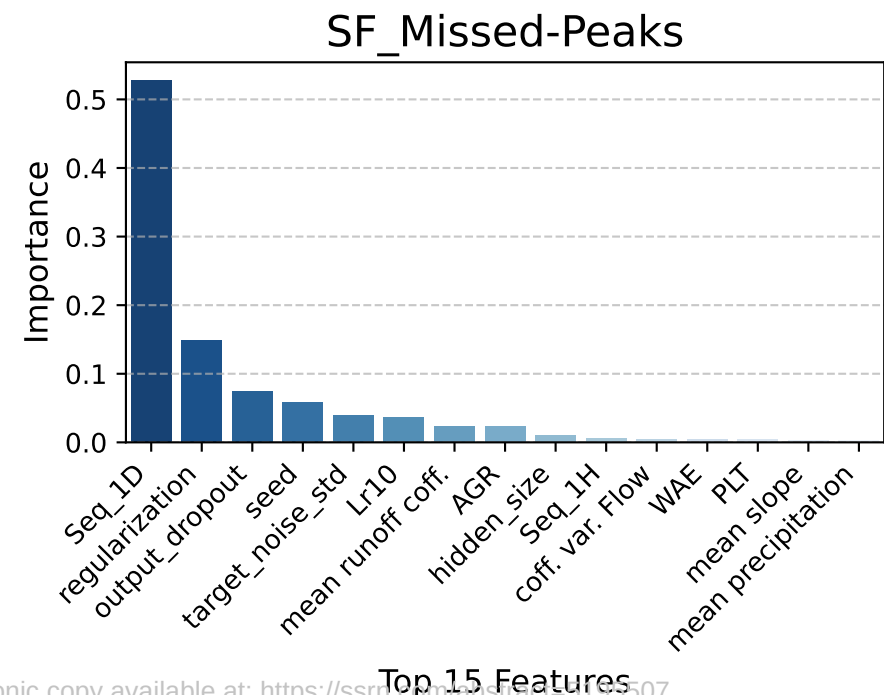
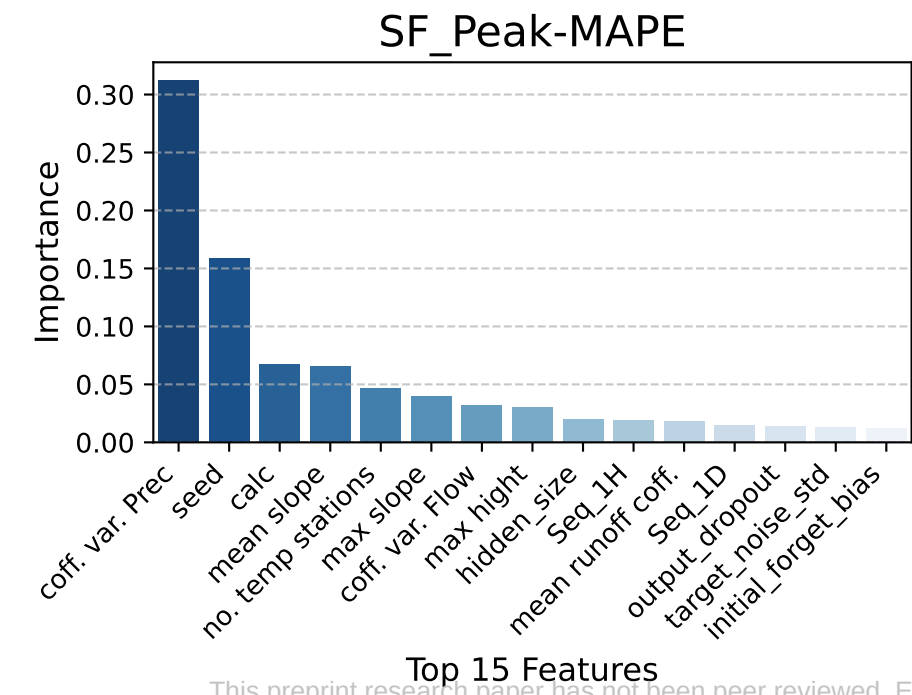
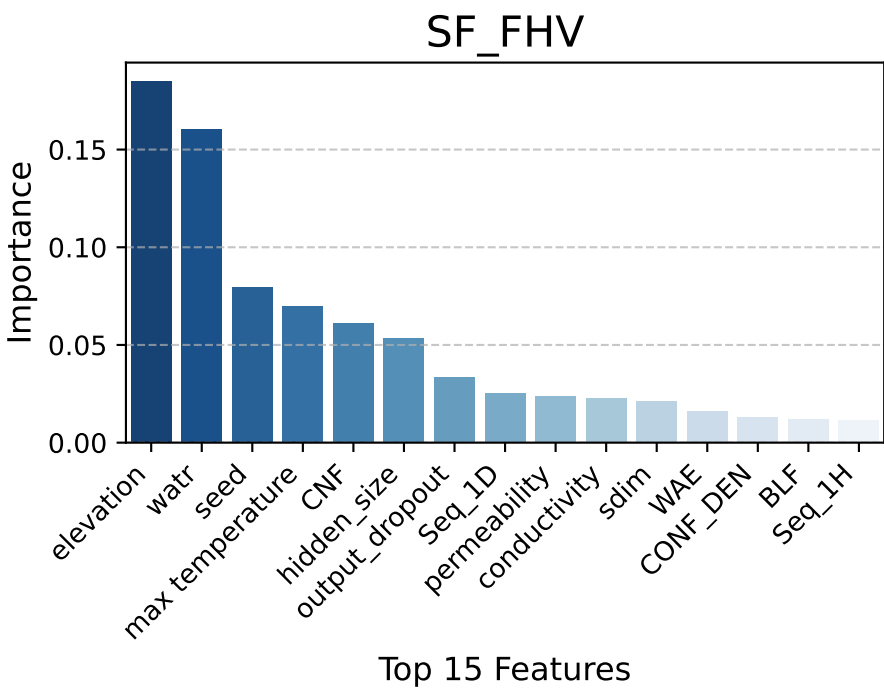
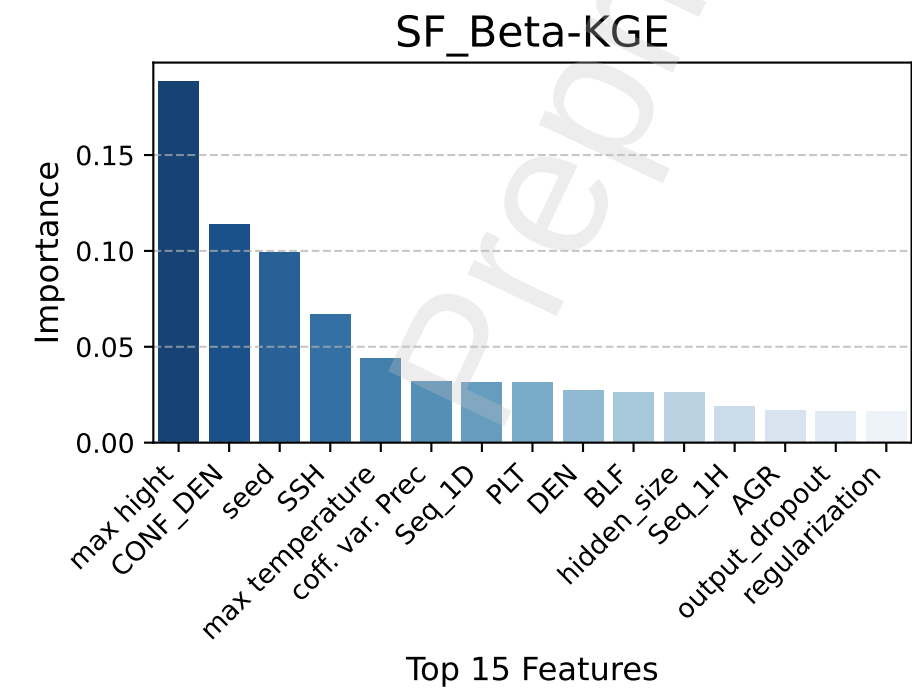
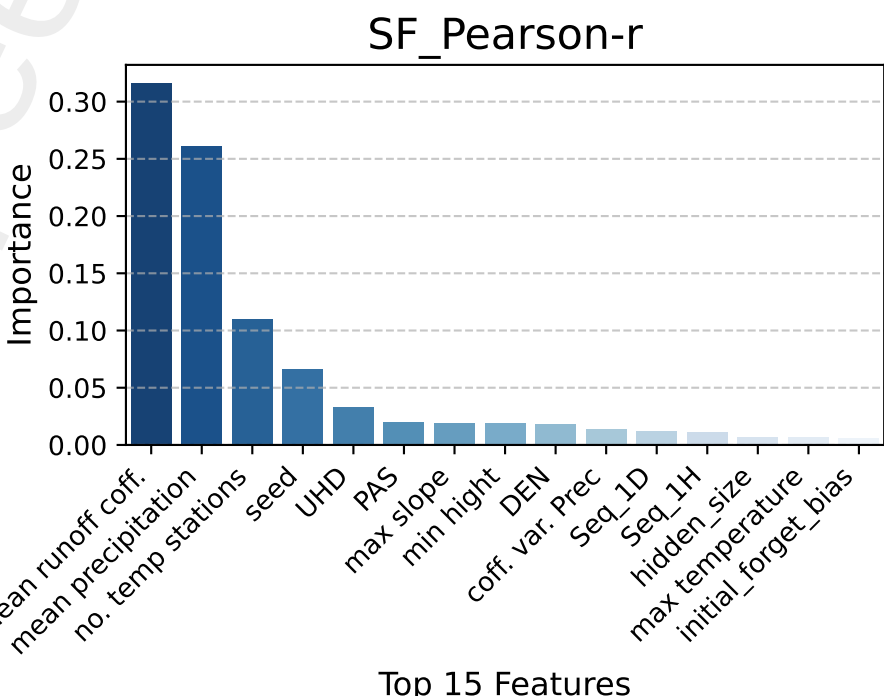
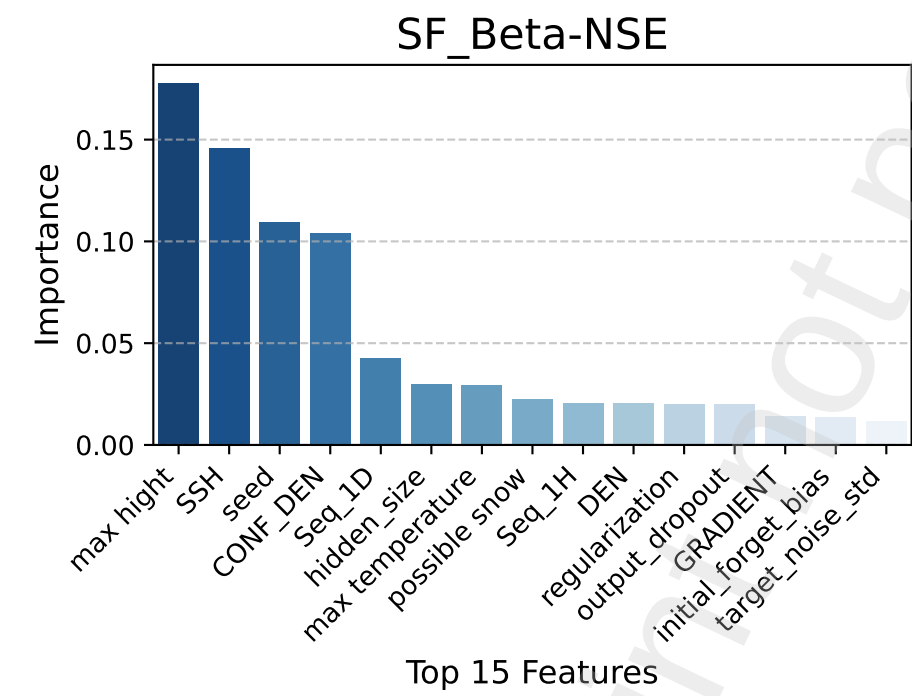
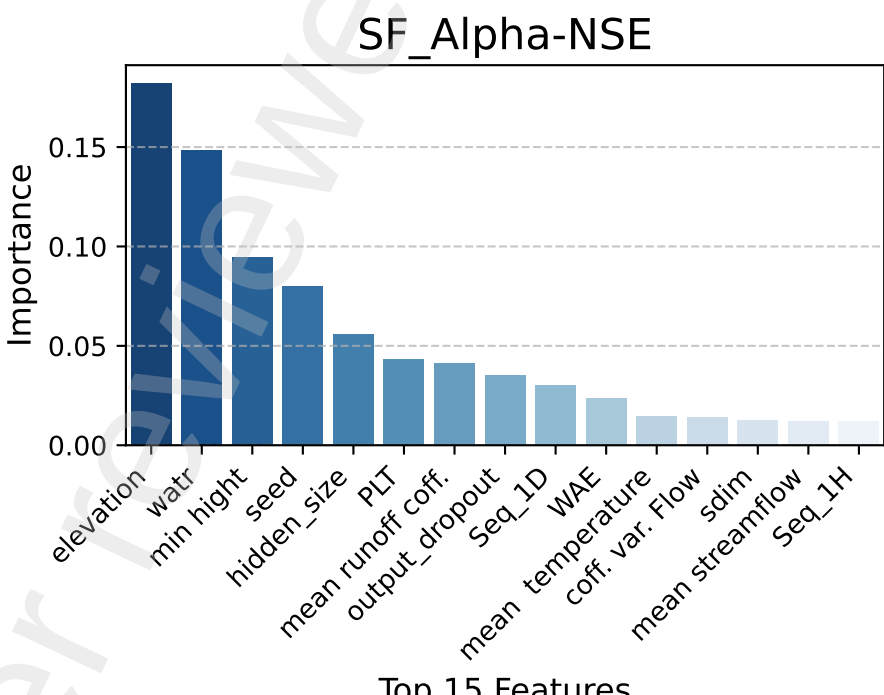
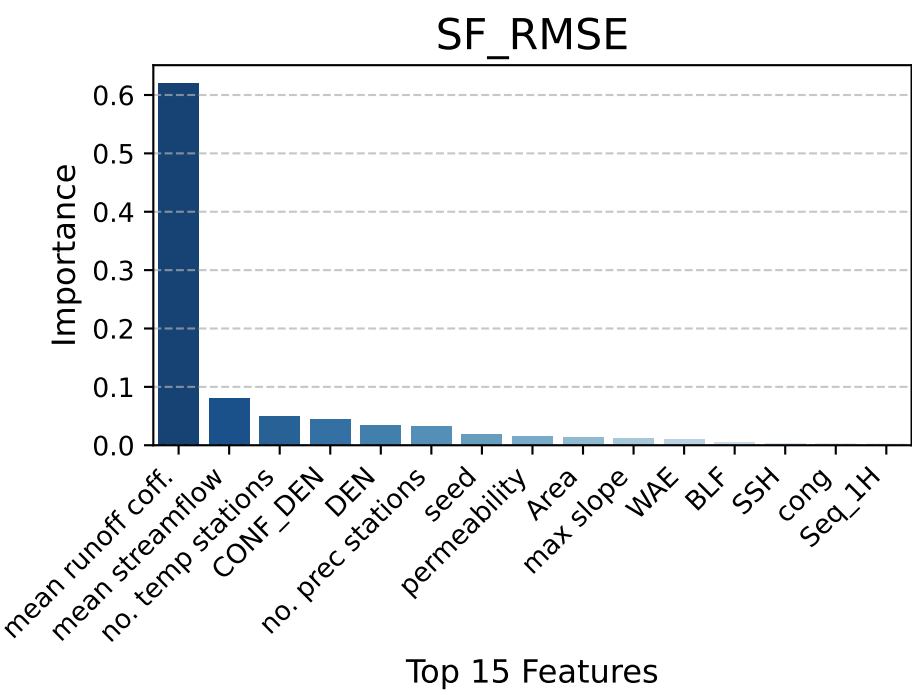
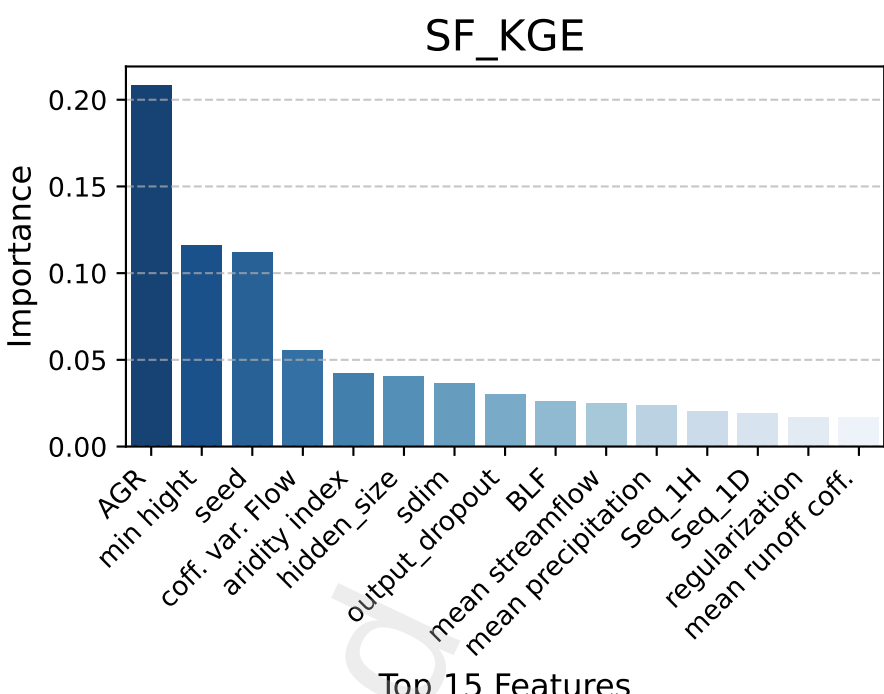
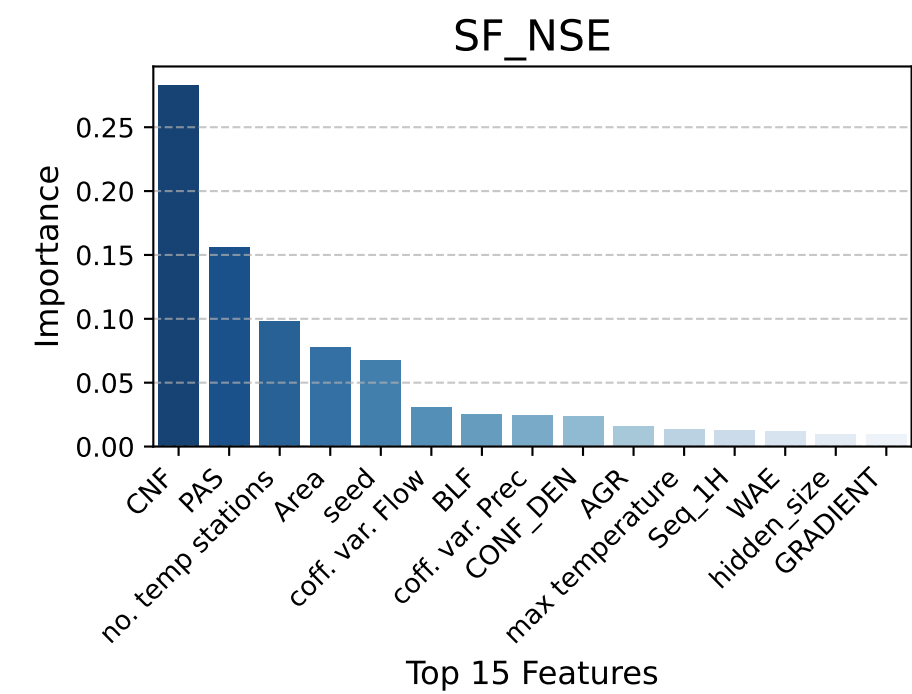


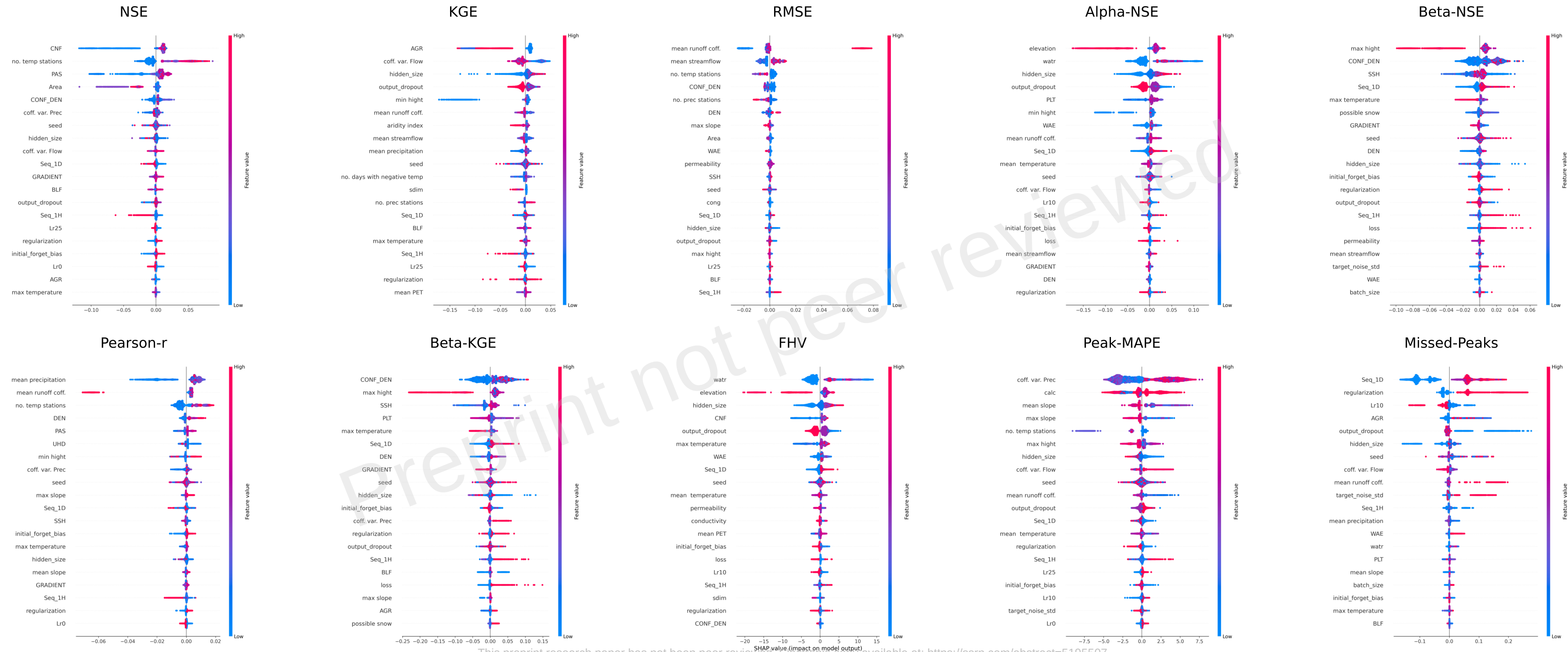
Correlation Heatmap: LSTMs' streamflow (SF) prediction metrics vs. catchment attributes on verification set



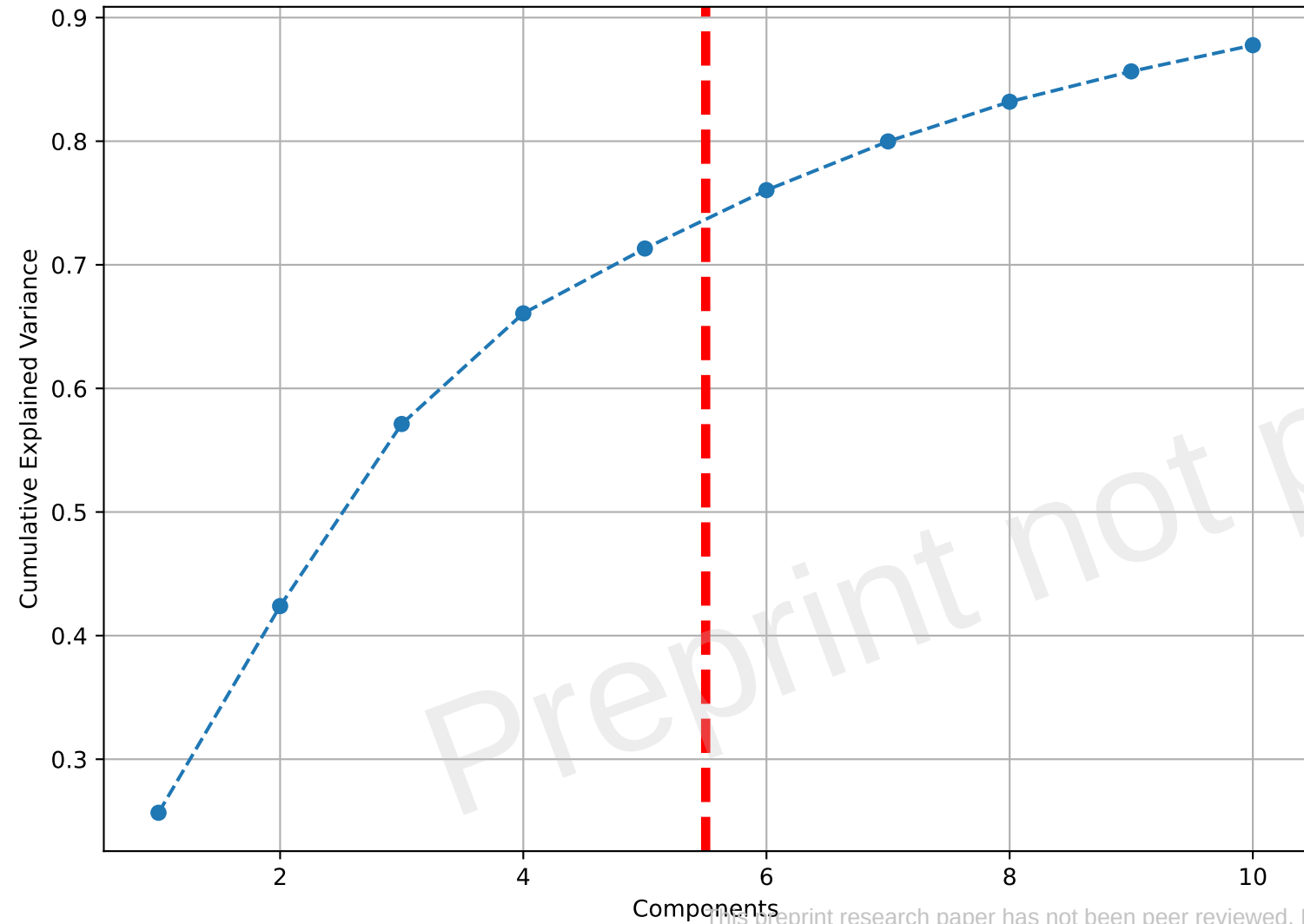


Feature Importance for Random Forest Model

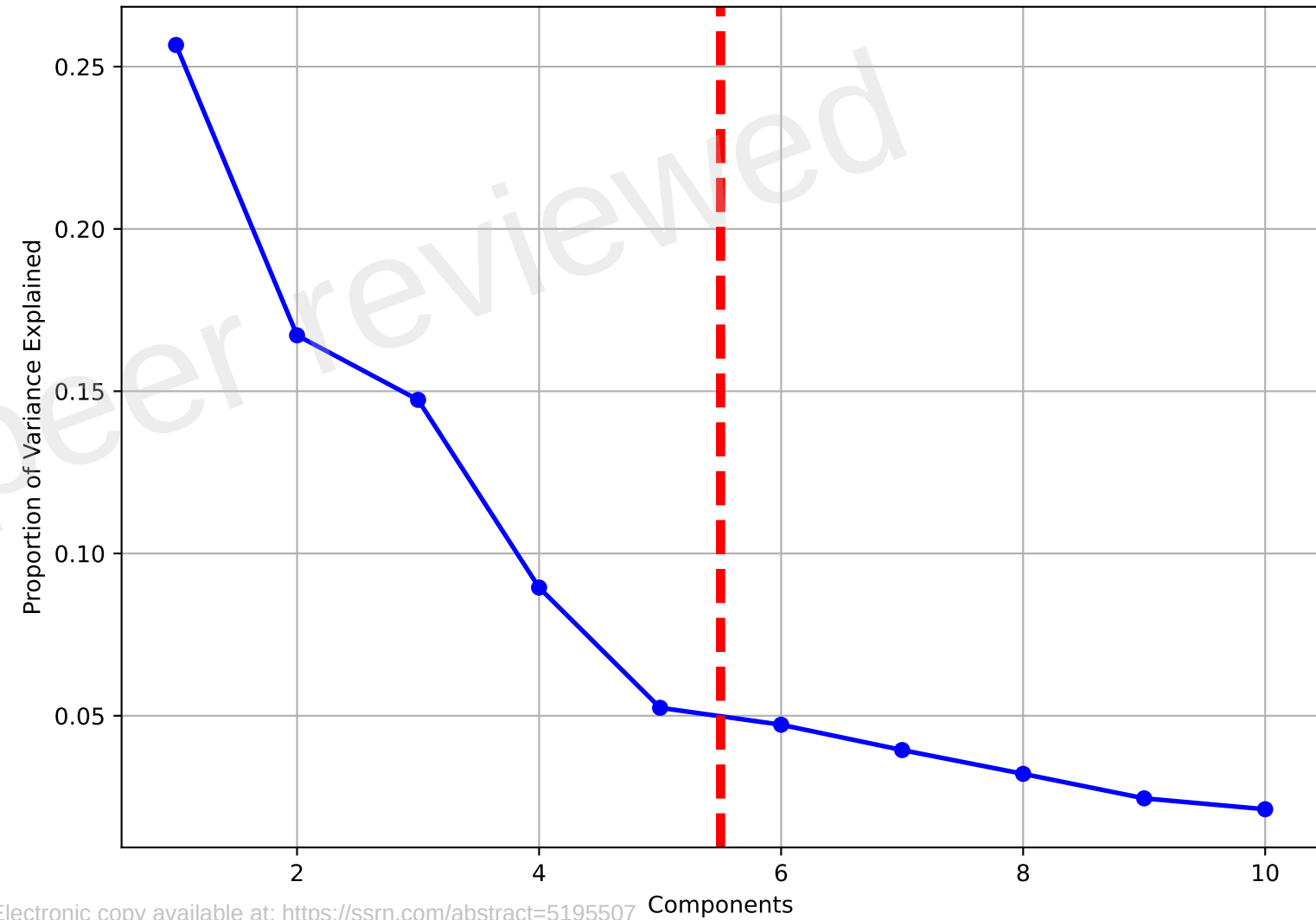




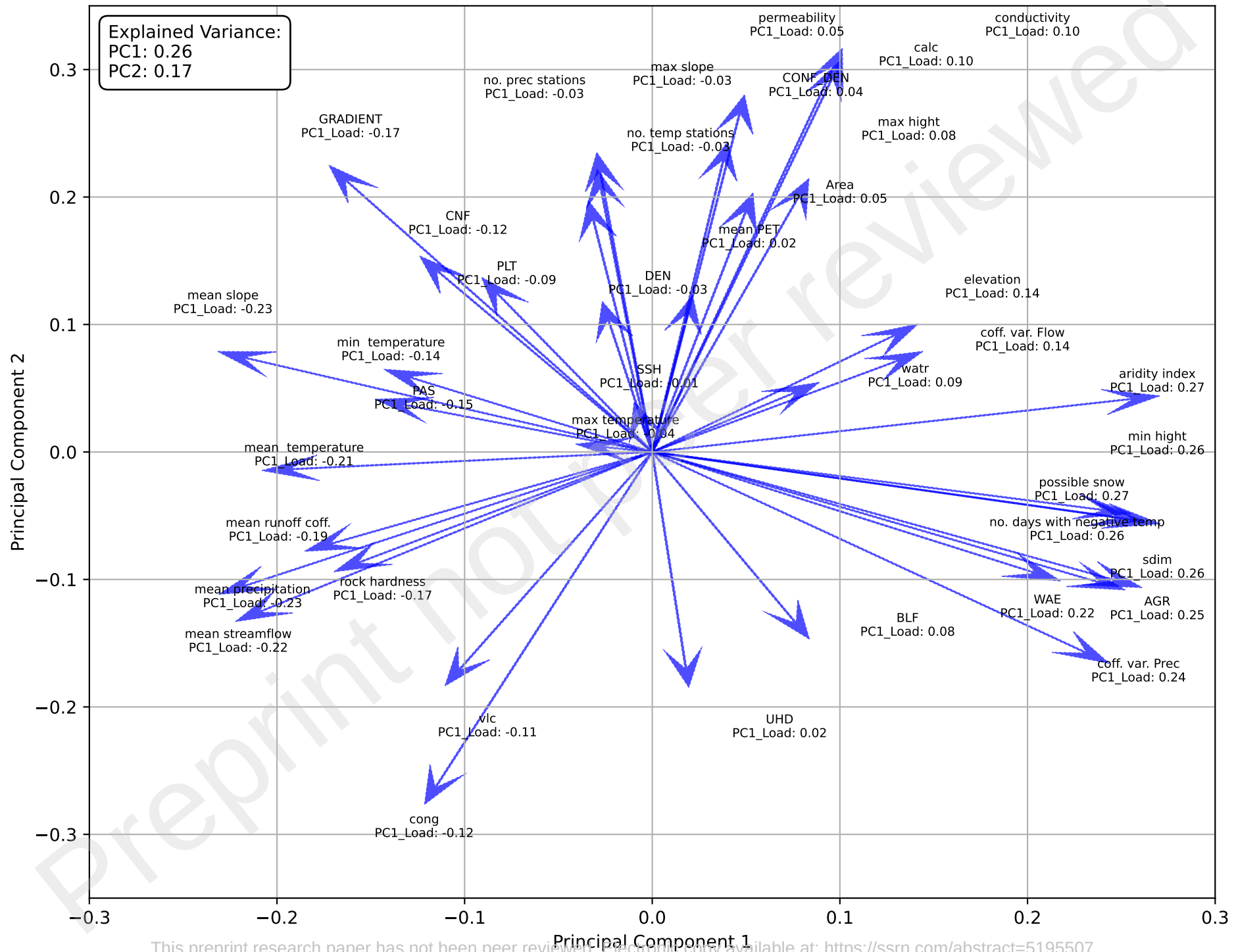
Cumulative Explained Variance



Scree Plot



Biplot of First Two Principal Components



Row	Station Name	Data Start	Data End	Area Km2	Annual PREC mm	Min Temp °C	Max Temp °C	Days with Negative Temp	RC	Aridity Index (AI)	Flow mm	Annual PET mm	GRAD %	Elevation		Land Use										Soil					COND	PERM	HARD
														MIN Height m	MAX Height m	UHD	AGR	PAS	BLF	CNF	PLT	SSH	WAE	DEN	CALC	CONG	SDIM	VLC	WATR	(1-5) class			
1	Abetxuko	9/3/2010	7/14/2019	679.7	1175	-9.91	39.29	225	0.33	1.31	241.31	968.09	25%	503	1549	3%	18%	13%	36%	7%	8%	14%	1%	1%	88%	0%	10%	0%	2%	4.6	3.0	2.8	
2	Abusu	10/1/2000	9/30/2021	1003.1	1176	-4.26	38.14	41	0.61	0.82	718.96	956.83	40%	13	1377	2%	0%	18%	18%	24%	13%	23%	0%	3%	97%	2%	0%	0%	1%	4.9	3.1	3.0	
3	Agautza	10/1/2000	9/30/2021	69.5	1537	-5.80	38.78	47	0.72	0.77	874.63	937.01	47%	184	1412	0%	0%	25%	50%	4%	13%	8%	0%	1%	100%	0%	0%	0%	0%	5.0	3.1	3.0	
4	Aitzu	10/1/2000	9/30/2021	56.8	1486	-5.90	37.57	51	0.63	0.67	886.67	951.77	45%	312	1431	1%	0%	21%	22%	31%	17%	5%	0%	3%	100%	0%	0%	0%	0%	5.0	3.1	3.0	
5	Aixola	7/1/1987	9/30/2019	4.8	1421	-17.30	38.13	40	0.44	0.65	619.60	929.90	44%	340	750	0%	0%	5%	5%	80%	10%	0%	0%	0%	100%	0%	0%	0%	0%	4.6	3.0	3.0	
6	Aizarnazabal	10/1/2000	9/30/2021	273.5	1653	-5.90	37.57	51	0.63	0.67	895.56	950.02	47%	20	1074	1%	1%	32%	14%	43%	4%	5%	0%	0%	97%	3%	0%	0%	0%	5.0	3.1	3.0	
7	Alegia	10/1/2000	10/2/2020	329.6	1416	-6.03	39.24	61	0.61	0.76	770.05	959.35	45%	92	1549	1%	0%	27%	30%	14%	18%	10%	0%	1%	100%	0%	0%	0%	0%	5.0	3.1	3.0	
8	Alegria	4/30/2010	2/7/2021	185.1	1074	-10.74	39.45	238	0.31	1.18	231.54	868.46	22%	508	1099	4%	36%	3%	42%	0%	5%	9%	1%	0%	81%	0%	19%	0%	0%	4.3	3.0	2.6	
9	Altzoa	10/1/2000	9/30/2021	460.3	1380	-5.34	39.28	44	0.60	0.74	775.15	955.84	44%	12	1363	2%	0%	26%	20%	25%	15%	11%	0%	1%	99%	0%	0%	0%	0%	5.0	3.1	3.0	
10	Amorebieta	10/1/2000	9/30/2021	233.4	1302	-5.72	40.12	51	0.71	0.73	912.70	950.78	43%	65	1330	2%	1%	24%	10%	27%	11%	17%	0%	8%	90%	10%	0%	0%	0%	4.7	2.9	2.9	
11	Anarbe	10/1/2000	9/30/2021	47.1	2031	-7.87	39.63	74	0.78	0.42	1744.02	951.18	54%	182	1052	0%	0%	18%	62%	5%	12%	2%	0%	0%	0%	87%	0%	12%	0%	1.9	2.9	4.0	
12	Araxes	1/5/2011	9/30/2021	93.1	1616	-7.00	38.69	69	0.75	0.50	1428.12	950.78	49%	119	1429	1%	0%	30%	41%	13%	11%	3%	0%	1%	100%	0%	0%	0%	0%	5.0	3.3	3.0	
13	Arenao	5/26/2005	9/1/2020	85.7	1200	-3.74	39.31	14	0.58	0.87	616.01	929.71	38%	45	821	1%	0%	28%	16%	20%	12%	20%	0%	4%	100%	0%	0%	0%	0%	5.0	3.1	3.0	
14	Areta	4/1/2013	9/30/2021	190.1	1149	-5.14	37.88	48	0.58	0.89	625.02	953.53	36%	118	1305	1%	0%	14%	37%	15%	9%	23%	0%	1%	100%	0%	0%	0%	0%	5.0	3.0	3.0	
15	Balmaseda	10/1/2000	9/30/2021	194.9	1064	-4.07	40.60	18	0.67	0.92	696.91	953.53	34%	172	1331	1%	3%	33%	36%	10%	7%	8%	0%	1%	90%	10%	0%	0%	0%	4.9	2.9	2.9	
16	Eibar	12/1/2013	9/30/2021	50.0	1451	-5.04	38.35	25	0.54	0.66	738.08	903.95	44%	94	812	5%	0%	34%	5%	41%	9%	5%	0%	0%	100%	0%	0%	0%	0%	4.6	3.0	3.0	
17	Elorrio	1/1/2001	9/30/2021	29.6	1368	-5.02	42.89	18	0.50	0.69	656.77	909.78	40%	167	1116	2%	1%	21%	4%	31%	21%	20%	0%	1%	100%	0%	0%	0%	0%	5.0	3.0	3.0	
18	Erenozu	10/1/2000	9/30/2021	215.8	1902	-6.43	38.56	36	0.72	0.45	1444.70	909.78	55%	23	1142	0%	0%	20%	41%	15%	15%	9%	0%	0%	2%	95%	0%	2%	1%	2.3	2.8	3.7	
19	Estanda	10/1/2000	9/30/2021	54.6	1352	-5.80	38.78	47	0.48	0.75	587.45	909.78	43%	164	966	2%	0%	26%	15%	32%	13%	11%	0%	0%	100%	0%	0%	0%	0%	5.0	3.0	3.0	
20	Etura	12/1/2012	9/30/2021	113.9	1201	-9.47	35.91	135	0.63	1.02	571.04	924.84	20%	548	1150	3%	37%	14%	39%	0%	1%	4%	0%	0%	90%	3%	7%	0%	0%	4.5	3.0	2.8	
21	Gardea	10/1/2000	9/30/2021	192.2	1025	-5.55	41.73	61	0.42	0.94	418.20	939.40	34%	134	1183	1%	1%	26%	27%	15%	8%	21%	0%	1%	98%	2%	0%	0%	0%	5.0	3.0	3.0	
22	Gatika	12/1/2013	9/30/2021	143.4	1308	-2.88	37.71	11	0.61	0.77	747.34	940.40	28%	5	687	6%	1%	38%	16%	20%	5%	13%	0%	1%	96%	4%	0%	0%	0%	4.8	3.0	3.0	
23	Herrerias	10/1/2000	9/30/2021	254.0	1103	-4.07	40.60	18	0.41	0.84	450.38	926.70	39%	50	1184	0%	0%	26%	14%	27%	13%	19%	0%	1%	100%	0%	0%	0%	0%	5.0	3.0	3.0	
24	Ibaiz Eder	10/1/2000	9/30/2021	65.4	1644	-5.90	37.57	51	0.51	0.62	738.76	895.34	49%	87	971	0%	0%	26%	27%	36%	3%	7%	0%	0%	100%	0%	0%	0%	0%	5.0	3.0	3.0	
25	Jaizubia	4/1/2013	9/30/2021	18.3	1988	-3.72	41.03	7	0.82	0.60	1244.23	902.12	36%	3	544	10%	0%	20%	33%	1%	9%	26%	0%	0%	52%	32%	0%	16%	0%	3.3	2.8	3.4	
26	Larrainazubi	4/1/2013	9/30/2021	19.1	1363	-3.55	39.57	4	0.74	1.01	702.34	953.67	23%	5	254	20%	0%	35%	29%	0%	0%	15%	0%	0%	12%	86%	2%	0%	0%	3.3	2.1	2.1	
27	Lasarte	10/1/2000	9/30/2021	791.3	1601	-5.77	38.84	50	0.69	0.62	971.40	874.21	45%	18	1549	1%	0%	28%	28%	21%	11%	9%	0%	1%	83%	17%	0%	0%	0%	4.7	3.1	3.0	
28	Leitzarain	10/1/2000	9/30/2021	114.2	1943	-7.34	38.69	69	0.75	0.47	1490.37	938.93	49%	49	1200	1%	0%	21%	22%	42%	4%	11%	0%	0%	17%	83%	0%	0%	0%	3.2	2.7	3.1	
29	Markina	12/1/2013	9/30/2021	34.0	1469	-5.46	37.80	28	0.61	0.71	823.09	955.75	45%	76	791	1%	0%	35%	3%	48%	5%	8%	0%	0%	100%	0%	0%	0%	0%	4.8	3.0	3.0	
30	Muxika	10/1/2000	9/30/2021	31.4	1351	-3.56	39.10	37	0.47	0.69	650.36	945.24	36%	11	625	0%	1%	25%	5%	43%	11%	14%	0%	0%	97%	3%	0%	0%	0%	4.2	3.0	3.0	
31	Oiarzun	10/1/2000	9/30/2020	55.9	2073	-5.49	38.57	27	0.81	0.46	1517.96	860.62	49%	6	831	1%	0%	19%	25%	9%	15%	26%	0%	4%	18%	38%	0%	44%	0%	2.1	2.5	3.8	
32	Onati	10/1/2000	9/30/2021	99.1	1442	-5.34	39.28	44	0.63	0.63	997.41	1005.28	47%	193	1362	1%	0%	21%	30%	19%	15%	12%	0%	2%	100%	0%	0%	0%	0%	5.0	3.0	3.0	
33	Oxandio	1/1/2003	9/30/2021	35.5	1361	-9.23	38.06	121	0.70	0.71	956.55	962.09	34%	549	1330	1%	0%	11%	40%	29%	7%	8%	0%	5%	100%	0%	0%	0%	0%	5.0	3.0	3.0	
34	Ozeta	4/1/2014	9/30/2021	97.5	1323	-8.58	36.86	90	0.58	1.04	540.12	961.37	29%	549	1547	3%	4%	24%	32%	0%	10%	25%	0%	1%	86%	0%	14%	0%	0%	4.6	3.0	2.7	
35	San Prudentzio	10/1/2000	9/30/2021	122.2	1206	-5.34	39.28	44	0.55	0.77	712.99	992.86	42%	171	1146	2%	0%	19%	23%	12%	26%	15%	0%	3%	99%	0%	0%	0%	1%	5.0	3.0	3.0	
36	Sangroniz	6/6/2005	9/30/2021	50.8	1281	-4.56	39.87	27	0.50	0.76	598.78	922.30	28%	5	475	11%	2%	26%	18%	16%	6%	19%	0%	1%	100%	0%	0%	0%	0%	4.6	3.0	3.0	
37	Saratxo	10/1/2000	9/29/2021	91.2	993	-5.07	41.45	37	0.46	0.94	440.65	910.58	35%	225	1140	1%	1%	27%	40%	5%	1%	23%	0%	2%	94%	6%	0%	0%	0%	4.9	2.9	2.9	
38	Sodupe	2/22/2001	8/29/2020	275.8	1118	-4.07	40.60	18	0.68	0.87	710.17	916.49	38%	49	717	1%	0%	27%	10%	28%	15%	19%	0%	4%	99%	1%	0%	0%	0%	5.0	3.0	3.0	
39	Urkizu	10/1/2000	9/30/2021	127.1	1178	-5.16	41.29	39	1.02	0.71	1311.44	912.66	42%	68	1377	1%	0%	16%	13%	31%	12%	23%	0%	4%	100%	0%	0%	0%	0%	5.0	3.2	3.0	
40	Zaratzam	1/1/2003	9/30/2021	512.3	1107	-4.78	39.28	47	0.57	0.88	607.34	940.69	38%	41	1305	1%	0%	17%	26%	19%	11%	24%	0%	1%	98%	1%	0%	0%	1%	5.0	3.1	3.0	

* PREC: precipitation; Temp: Temperature; RC: Runoff Coefficient; PET: Potential Evapotranspiration; GRAD: Gradient (Slope)

* Land Use Distribution: Urban (UHD), Agriculture (AGR), Pasture (PAS), Broadleaf Forest (BLF), Coniferous Forest (CNF), Plantation (PLT), Shrublands (SSH), Water bodies (WAE)

* Soil Composition: CALC: calcareous soils; CONG: conglomerate soils; SDIM: sedimentary soils; VLC: volcanic soils; WATR: wetlands and water associated ecosystems

* Soil Composition class: COND (soil conductivity), PERM (permeability), and HARD (hardness)

Attribute
Area
CONF_DEN
GRADIENT
max slope
mean slope
elevation
min height
max height
UHD
AGR
PAS
BLF
CNF
PLT
SSH
WAE
DEN
calc
cong
sdim
vlc
watr
conductivity
permeability
rock hardness
no. prec stations
no. temp stations
possible snow
no. days with negative temp
mean runoff coeff.
aridity index
mean precipitation
mean streamflow
mean temperature
min temperature
max temperature
Coeff. var. Prec
Coeff. var. Flow
mean PET

Definition
Contributing area to the downstream end of the segment
Number of rivers confluences by catchment area
Mean gradient through the reach (vertical change/horizontal length)
max slope of catchment
average slope of catchment
Average catchment elevation upstream the river reach
min catchment elevation
max catchment elevation
Surface occupied by urban areas upstream the river reach
Surface occupied by agricultural land upstream the river reach
Surface occupied by pasture upstream the river reach
Surface occupied by broadleaf forest upstream the river reach
Surface occupied by coniferous forest upstream the river reach
Surface occupied by plantations upstream the river reach
Surface occupied by moors, heathland, scrub and shrubs upstream the river reach
Surface occupied by wetlands and water ecosystems upstream the river reach
Surface occupied by denuded areas upstream the river reach
Area occupied by calcareous rocks upstream the river reach
Area occupied by conglomerate rocks upstream the river reach
Area occupied by sedimentary rocks upstream the river reach
Area occupied by volcanic rocks upstream the river reach
Area occupied by wetlands and water associated ecosystems upstream the river reach
Average soil conductivity upstream the river reach (derived from geology variables). Reaches with MN_watr and MN_othe=1 this value have 0 for this field
Average terrain permeability upstream the river reach (derived from geology variables). Reaches with MN_watr and MN_othe=1 this value have 0 for this field
Average soil hardness upstream the river reach (derived from geology variables). Reaches with MN_watr and MN_othe=1 this value have 0 for this field
Number of stations participated in calculating lumped prec values for basins
Number of stations participated in calculating lumped temp values for basins
Percentage of number of days with negative temp on total number of days
Number of days with negative temp in the dataset
yearly average runoff coefficient
Aridity Index
yearly average precipitation
yearly average streamflow
yearly average temperature
yearly min temperature
yearly max temperature
Coefficient of variation of precipitation
Coefficient of variation of streamflow
average potential evapotranspiration

Group	Units/Values
Topography	km ²
Topography	Number/km ²
Topography	ratio
Topography	°
Topography	°
Topography	m
Topography	m
Topography	m
Land Uses	%
Land Uses	%
Land Uses	%
Land Uses	%
Land Uses	%
Land Uses	%
Land Uses	%
Land Uses	%
Land Uses	%
Land Uses	%
Geology	%
Geology	%
Geology	%
Geology	%
Geology	%
Geology	Class: 1 - 5
Geology	Class: 1 - 5
Geology	Class: 1 - 5
Hydrology	Number
Hydrology	Number
Hydrology	%
Hydrology	Number
Hydrology	dimensionless
Hydrology	dimensionless
Hydrology	mm
Hydrology	mm
Hydrology	°C
Hydrology	°C
Hydrology	°C
Hydrology	dimensionless
Hydrology	dimensionless
Hydrology	mm

Hyperparameter		Range
hidden size		16, 32, 64, 128, 256
batch size		32, 64, 128, 256
output dropout		0, 0.2, 0.4
initial forget bias		-3, -1, 0, 1, 3
learning rates	Lr0	1e-3, 1e-2, 5e-2
	Lr10	5e-4, 1e-3, 5e-3
	Lr25	1e-4, 1e-3
target noise std		0, 0.01, 0.02, 0.05, 0.1
loss function		NSE, RMSE
seq length daily		146, 182, 365, 730, 1095
seq length hourly		168, 336, 504, 672, 1344, 2016, 4032, 6720, 8064, 8760
regularization		tie_frequencies, None

Metric	Attribute	Correlation
SF_RMSE	mean runoff coeff.	0.70
SF_MSE	mean runoff coeff.	0.63
SF_RMSE	mean streamflow	0.58
SF_MSE	mean streamflow	0.50
SF_NSE	GRADIENT	0.47
SF_Pearson-r	GRADIENT	0.44
SF_KGE	mean slope	0.42
SF_Peak-MAPE	coeff. var. Prec	0.41
SF_NSE	mean slope	0.40
SF_Pearson-r	max slope	0.39
SF_Pearson-r	mean slope	0.38
SF_KGE	mean precipitation	0.38
SF_NSE	max slope	0.38
SF_RMSE	mean precipitation	0.38
SF_Peak-MAPE	aridity index	0.35
SF_KGE	GRADIENT	0.34
SF_KGE	CNF	0.33
SF_NSE	CNF	0.32
SF_FLV	watr	-0.31
SF_RMSE	max slope	-0.31
SF_RMSE	WAE	-0.31

Metric	Attribute
SF_MSE	max slope
SF_NSE	min hight
SF_MSE	aridity index
SF_NSE	AGR
SF_Pearson-r	aridity index
SF_RMSE	CONF_DEN
SF_Peak-MAPE	mean precipitation
SF_Peak-MAPE	GRADIENT
SF_Pearson-r	sdim
SF_KGE	possible snow
SF_NSE	aridity index
SF_RMSE	Area
SF_Peak-MAPE	mean slope
SF_NSE	sdim
SF_RMSE	aridity index
SF_KGE	aridity index
SF_KGE	min hight
SF_KGE	coeff. var. Prec
SF_KGE	sdim
SF_Pearson-r	coeff. var. Prec
SF_NSE	coeff. var. Prec

Correlation

-0.32

-0.32

-0.32

-0.32

-0.33

-0.34

-0.34

-0.34

-0.34

-0.35

-0.36

-0.37

-0.38

-0.39

-0.40

-0.44

-0.45

-0.45

-0.47

-0.49

-0.51

Attributes	Principal components						
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Area	0.054	-0.203	-0.099	0.355	-0.247	-0.061	-0.053
CONF_DEN	0.040	-0.243	-0.068	0.344	-0.157	-0.054	-0.019
GRADIENT	-0.172	-0.225	-0.096	-0.106	-0.011	-0.018	-0.048
max slope	-0.030	-0.235	-0.108	0.118	0.188	0.247	-0.020
mean slope	-0.232	-0.079	-0.198	-0.140	-0.094	-0.079	-0.137
elevation	0.142	-0.100	-0.295	-0.112	0.110	0.169	0.008
min hight	0.255	0.048	-0.133	-0.126	0.119	0.107	0.117
max hight	0.083	-0.214	-0.285	0.016	0.084	0.157	-0.011
UHD	0.019	0.185	0.248	0.158	-0.119	0.190	0.076
AGR	0.252	0.108	-0.053	0.008	-0.082	-0.120	0.101
PAS	-0.148	-0.041	0.222	-0.121	-0.214	0.191	-0.056
BLF	0.084	0.147	-0.229	0.091	0.067	0.306	-0.086
CNF	-0.124	-0.154	0.053	-0.247	-0.222	-0.262	0.056
PLT	-0.091	-0.137	-0.111	0.036	0.278	-0.148	-0.272
SSH	-0.010	-0.039	0.151	0.298	0.390	-0.124	0.065
WAE	0.218	0.101	-0.090	0.044	-0.091	-0.248	-0.176
DEN	-0.027	-0.118	-0.068	0.032	0.337	0.028	0.451
calc	0.100	-0.307	0.123	-0.178	0.068	-0.057	0.037
cong	-0.121	0.276	-0.102	0.145	-0.139	0.145	-0.095
sdim	0.261	0.106	-0.050	0.035	-0.031	-0.093	0.056
vlc	-0.110	0.183	-0.094	0.159	0.241	-0.237	0.176
watr	0.089	-0.055	-0.118	0.250	-0.061	0.028	-0.378
conductivity	0.101	-0.317	0.128	-0.153	0.031	0.077	0.023
permeability	0.049	-0.280	-0.114	-0.175	0.044	-0.113	-0.123
rock hardness	-0.170	0.094	-0.228	0.006	0.150	-0.244	-0.124
no. prec stations	-0.029	-0.222	-0.042	0.279	-0.147	-0.034	0.172
no. temp stations	-0.034	-0.197	-0.078	0.326	-0.220	-0.094	0.224
possible snow	0.271	0.056	-0.166	-0.002	-0.071	-0.075	0.021
no. days with negative temp	0.262	0.054	-0.186	-0.009	-0.072	-0.093	0.024
mean runoff coff.	-0.185	0.078	-0.116	0.078	0.069	0.207	0.321
aridity index	0.270	-0.044	0.143	0.124	0.036	0.098	-0.037
mean precipitation	-0.231	0.112	-0.218	-0.094	-0.053	-0.044	-0.046
mean streamflow	-0.222	0.133	-0.222	-0.001	0.015	0.051	0.112
mean temperature	-0.208	0.015	0.232	0.147	-0.011	0.014	-0.126
min temperature	-0.143	-0.065	0.187	0.163	0.070	0.080	-0.076
max temperature	-0.041	-0.006	0.114	0.127	0.366	-0.066	-0.350
coff. var. Prec	0.243	0.166	0.136	-0.007	0.119	0.050	-0.151
coff. var. Flow	0.144	-0.079	0.263	-0.054	0.105	-0.127	0.117
mean PET	0.021	-0.123	-0.027	-0.059	-0.006	0.483	-0.140
Explained Variance Ratio	25.7%	16.7%	14.7%	8.9%	5.2%	4.7%	3.9%
Cumulative Variance	25.7%	42.4%	57.1%	66.1%	71.3%	76.0%	80.0%

PC8	PC9	PC10
-0.024	0.066	-0.036
-0.127	0.058	-0.100
0.084	0.255	0.012
0.046	0.437	-0.144
0.036	0.006	0.007
0.109	0.064	-0.037
-0.027	-0.128	-0.026
0.092	0.111	0.057
-0.052	-0.050	0.279
0.143	-0.005	0.026
0.088	0.153	-0.151
0.318	-0.117	-0.233
-0.297	0.122	0.117
-0.103	-0.288	0.324
-0.119	0.060	-0.094
0.107	0.069	0.258
-0.326	0.110	0.153
0.101	-0.131	-0.016
-0.118	0.149	0.079
0.067	-0.023	0.123
-0.001	0.009	-0.276
-0.389	-0.130	-0.114
0.111	-0.017	0.079
0.183	-0.292	0.031
-0.015	-0.166	-0.322
0.203	-0.141	0.099
0.090	-0.121	0.098
-0.090	-0.006	0.019
-0.052	0.010	0.078
0.112	-0.211	0.292
-0.009	0.042	-0.046
-0.069	-0.036	0.025
0.017	-0.117	0.152
0.162	-0.131	0.050
0.206	-0.270	-0.126
0.142	0.250	0.386
0.040	0.013	0.051
-0.120	-0.208	-0.242
-0.425	-0.290	0.045
3.2%	2.5%	2.1%
83.2%	85.6%	87.8%

1 *Figure 1. Study area, including 40 catchments of Basque Country in north of Spain*

2 *Figure 2. Methodology for Assessing relations between catchments' attributes and LSTMs' test performances*

3 *Figure 3. Correlation heatmap showing the relationships between catchment attributes and streamflow (SF) test metrics. Darker*
4 *colors represent stronger correlations, with both positive and negative relationships between various features and metrics.*

5 *Figure 3. Random Forest prediction accuracy for different test performance metrics for streamflow. The Random Forest was*
6 *trained on Hyperparameters, and Attributes as inputs and the metrics as outputs. The figure suggests that a Random Forest model*
7 *can be trained in a way that accurately predicts the performance outcomes of a configured LSTMs in every catchment by knowing*
8 *their attributes.*

9 *Figure 4. Feature importance ranking for 15 top features for different target metrics derived from the trained Random Forest*
10 *model. This figure highlights the relative importance of various features in predicting the test metrics, emphasizing the most*
11 *influential attributes shaping DNNs' accuracy.*

12 *Figure 5. SHAP summary plots for the 20 most influential catchment attributes and LSTM hyperparameters affecting regional*
13 *model performance across Basque Country catchments. Each dot represents a single model-catchment instance, with color*
14 *gradients showing feature magnitudes (red: high, blue: low). The SHAP value axis indicates feature positive and negative*
15 *contribution to test performance metrics. Aggregated insights highlight dominant hydrological and architectural drivers of LSTM*
16 *performance across diverse evaluation criteria.*

17 *Figure 6. PCA results applied to the test_DATASET, showing the distribution of principal components. This plot illustrates how*
18 *much variability is captured by each component, providing insights into the test_DATASET's structure.*

19 *Figure 7. Biplot of the PCA analysis. This figure displays the principal components in relation to the original features, visually*
20 *representing how each feature contributes to the principal components and their interactions.*

- 1 *Table 1. A brief summary of 40 URA catchments' attributes in the case study.*
- 2 *Table 2. Definitions of the catchments' attributes employed in this study.*
- 3 *Table 3. The initial defined hyperparameters space designed for random search*
- 4 *Table 4. The attributes with correlations beyond the ± 0.3 threshold, linking them to their influence on LSTM performance.*
- 5 *Table 5. PCA components' loads and the explained variance ratios*
- 6
- 7